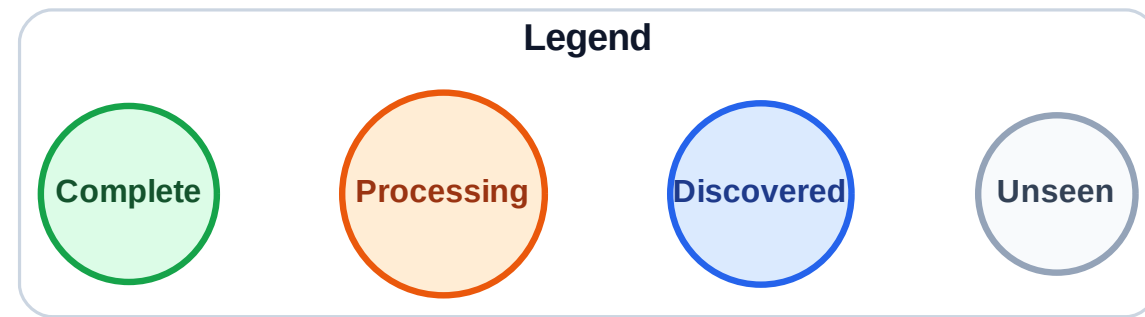


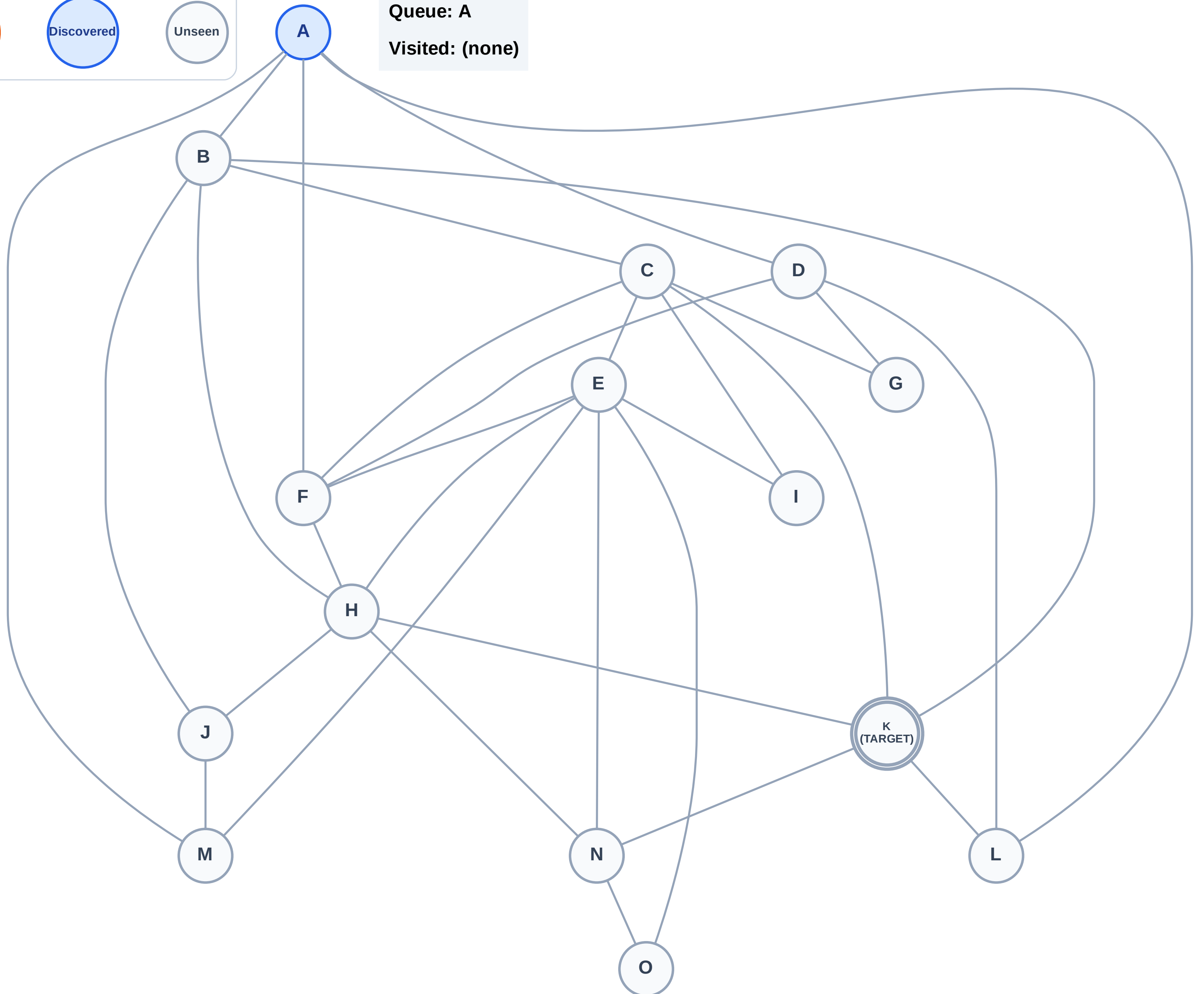
BFS Traversal

Step 1: Start at A (Seeking target node K)



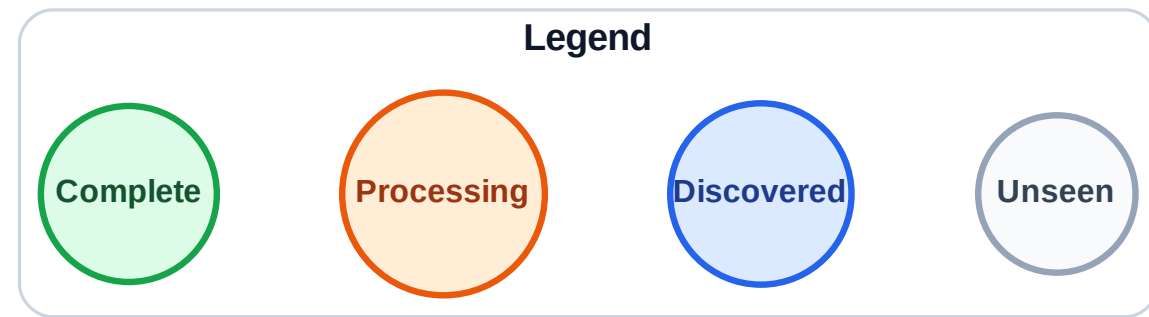
Queue: A

Visited: (none)

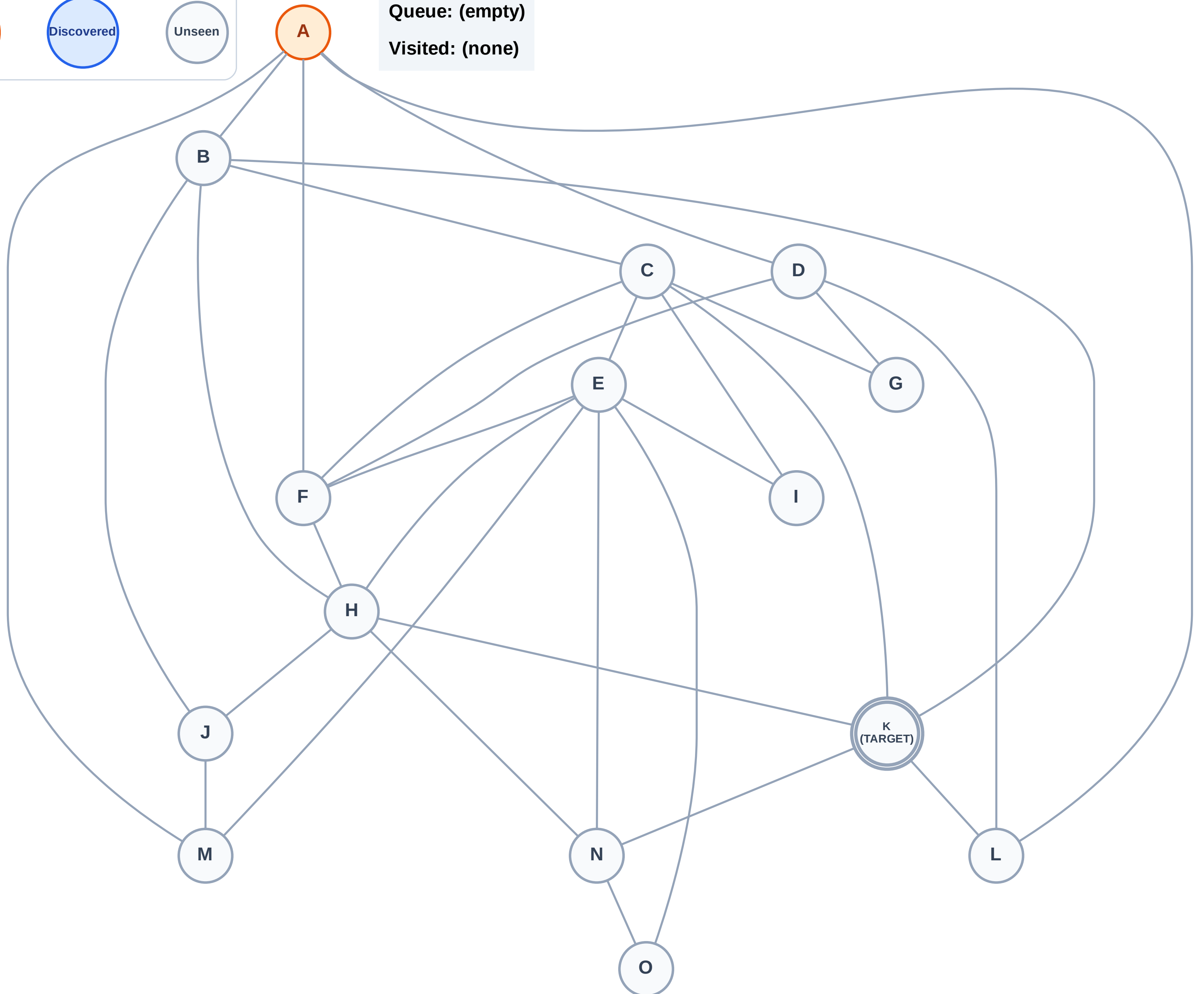


BFS Traversal

Step 2: Process node A

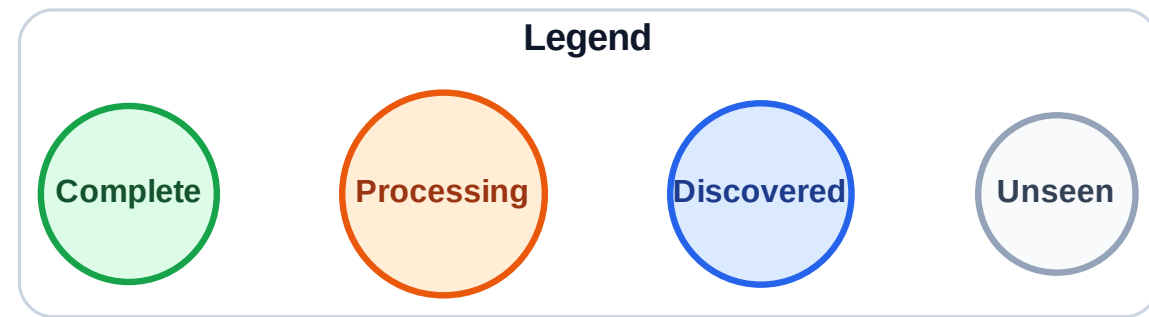


Queue: (empty)
Visited: (none)



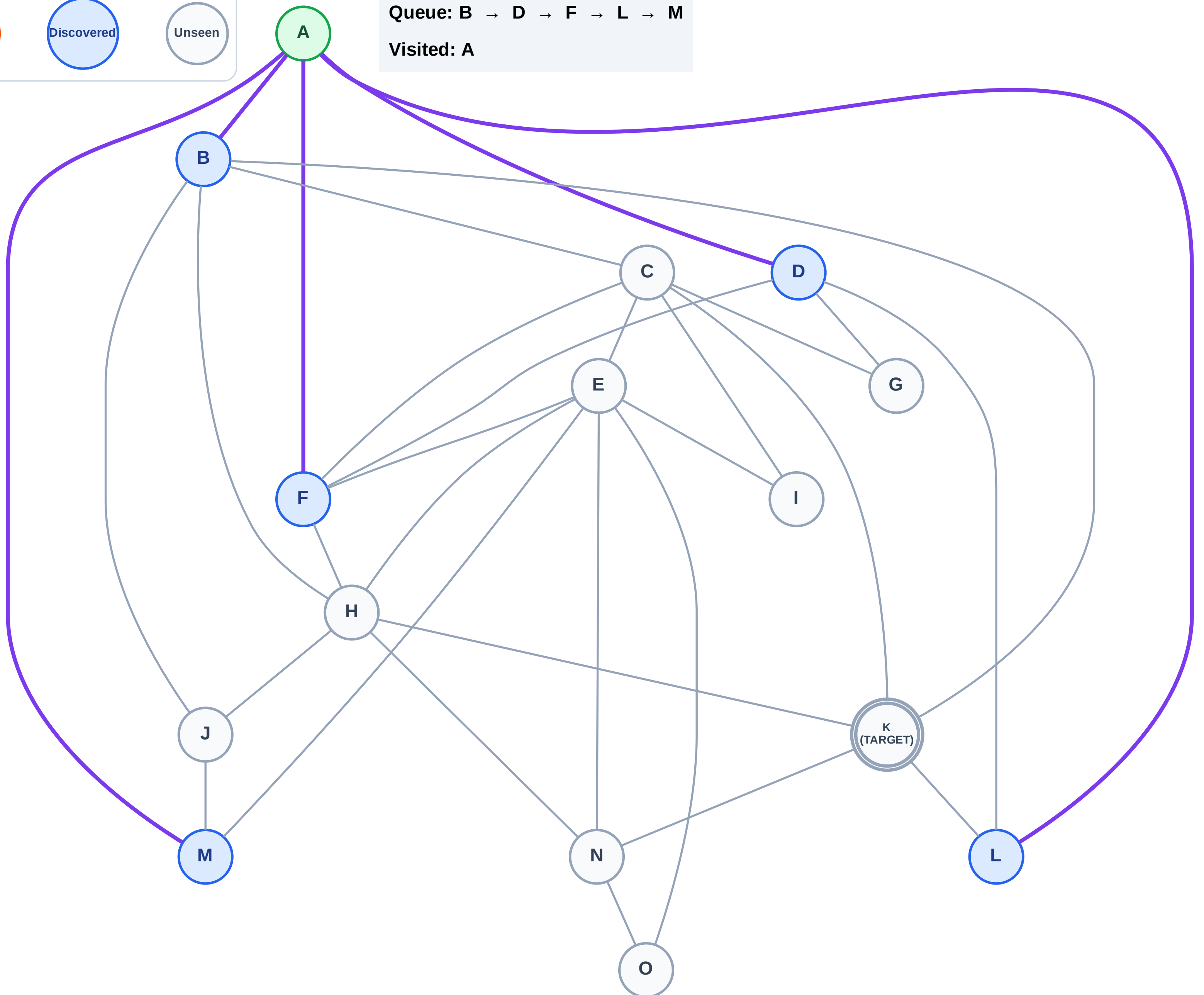
BFS Traversal

Step 3: Discover B, D, F, L, M from A



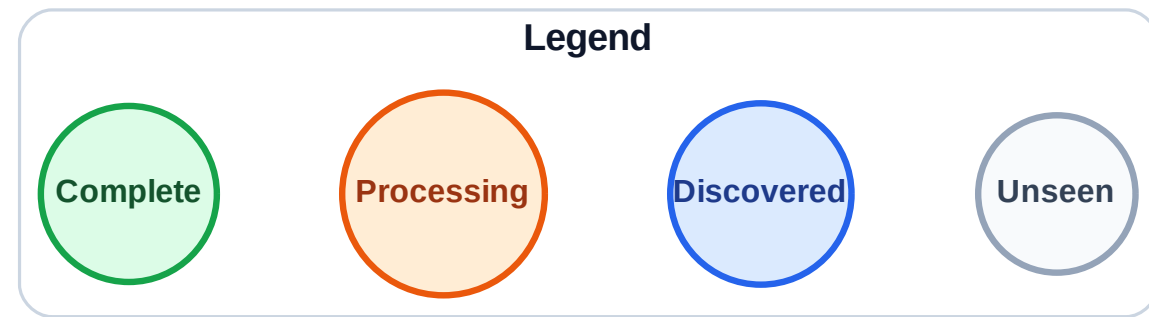
Queue: B → D → F → L → M

Visited: A



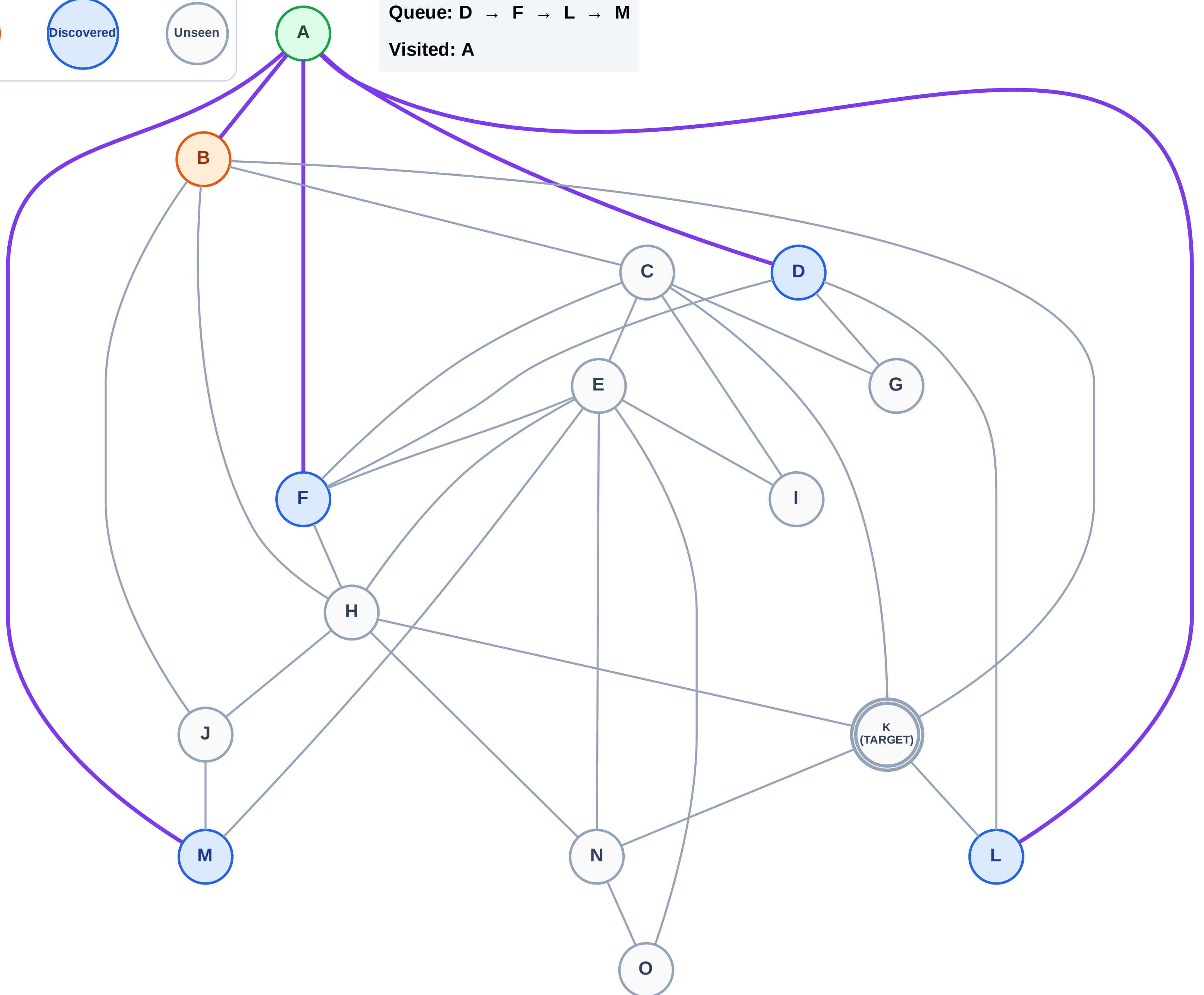
BFS Traversal

Step 4: Process node B



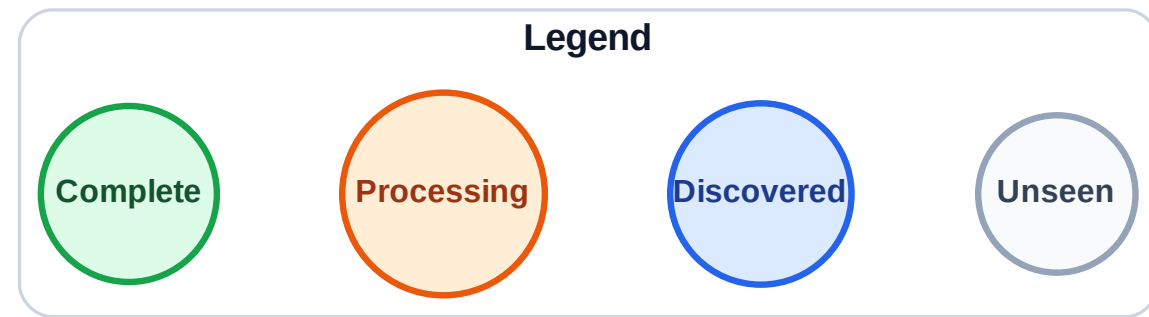
Queue: D → F → L → M

Visited: A

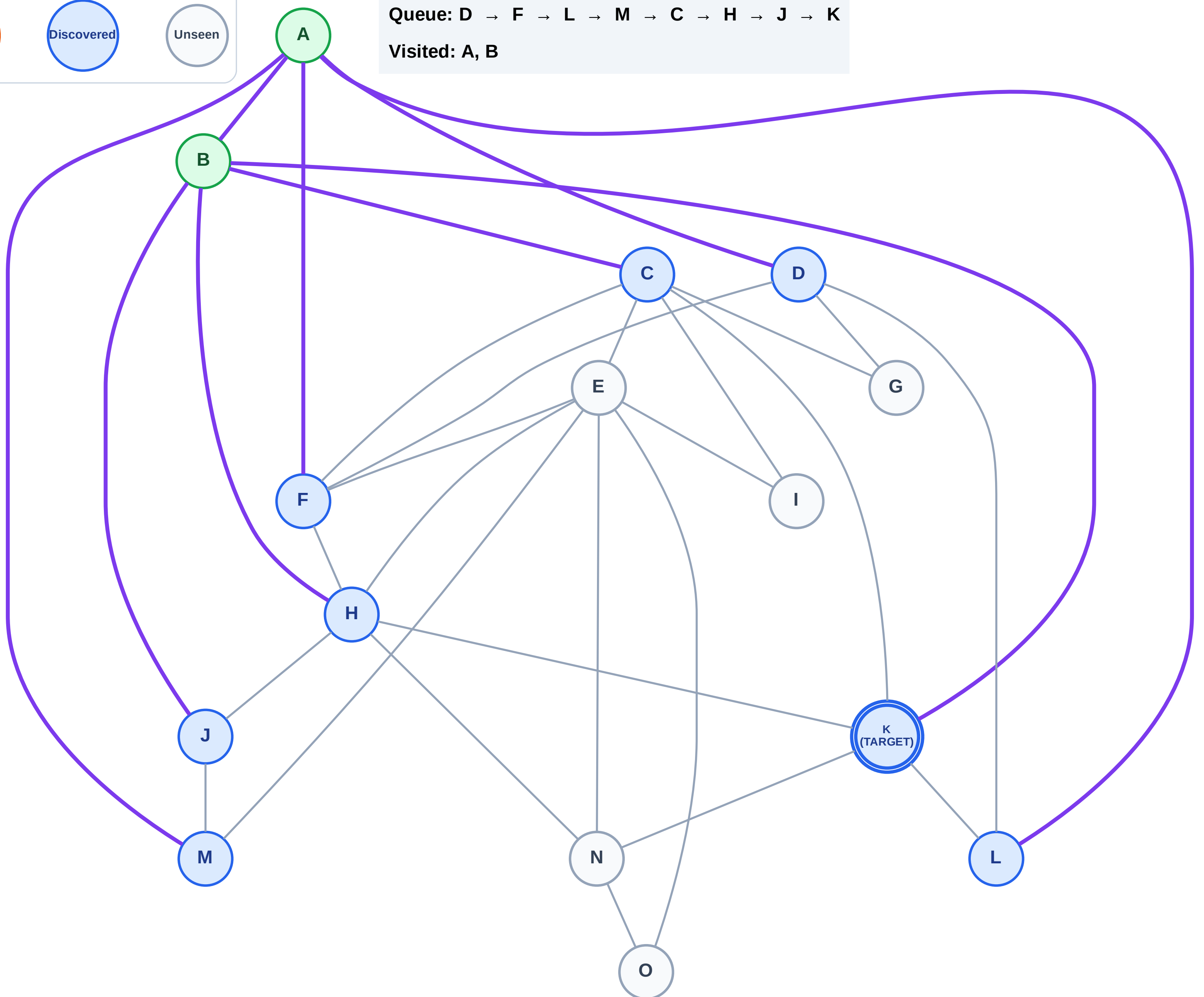


BFS Traversal

Step 5: Discover C, H, J, K from B

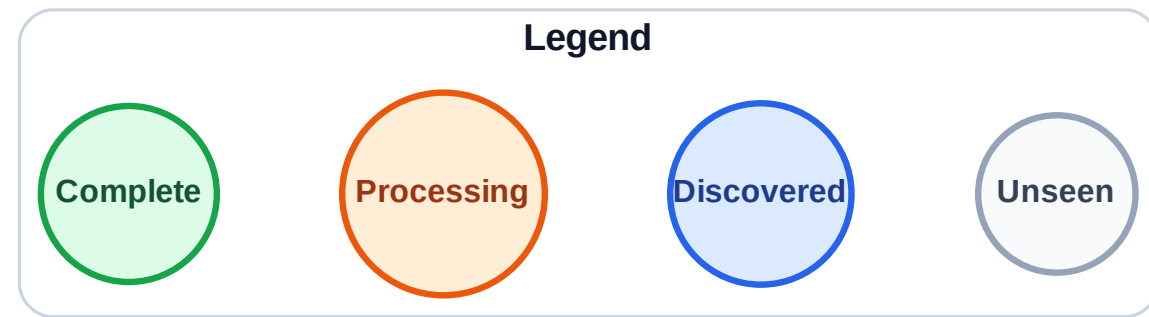


Queue: D → F → L → M → C → H → J → K
Visited: A, B



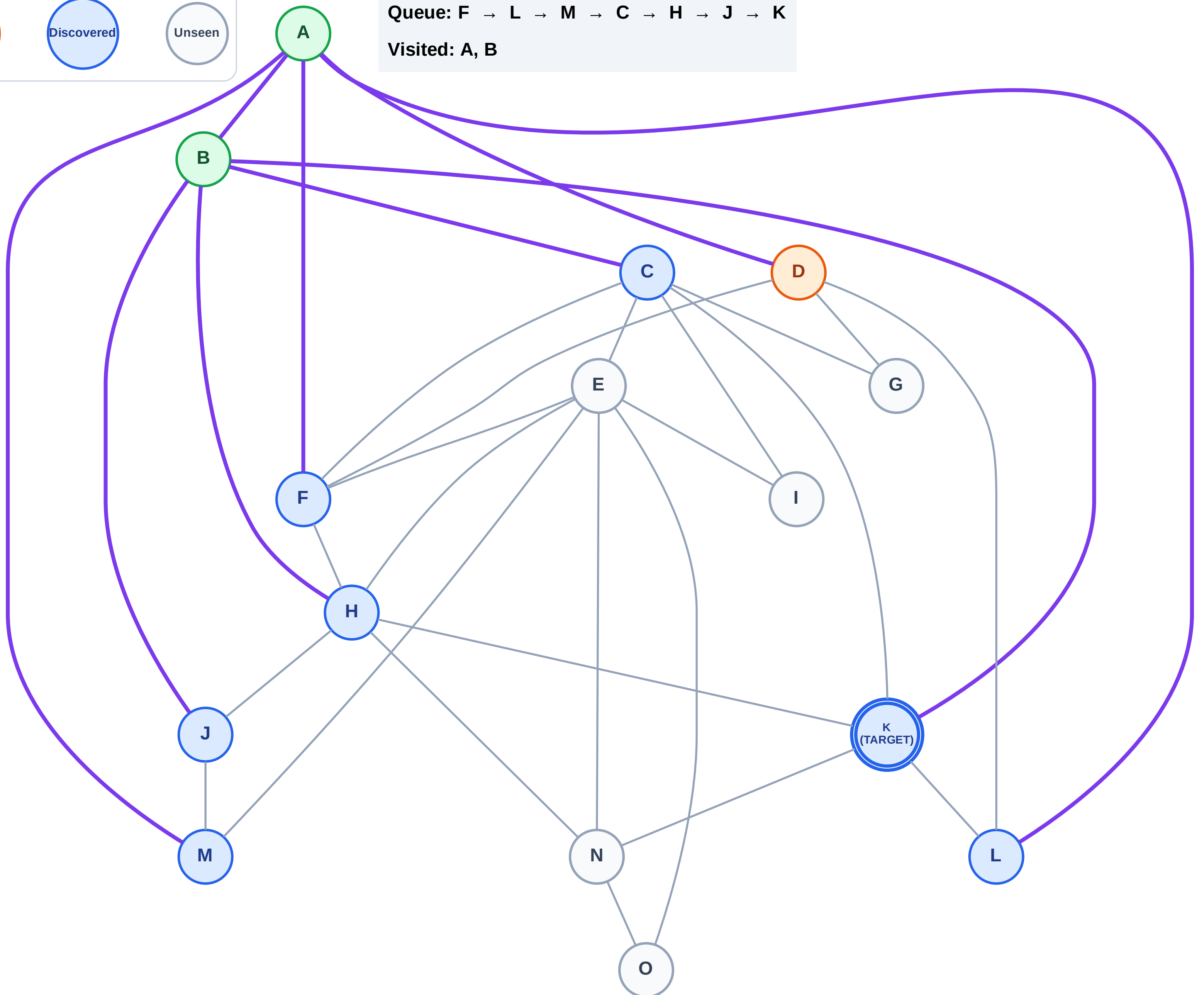
BFS Traversal

Step 6: Process node D



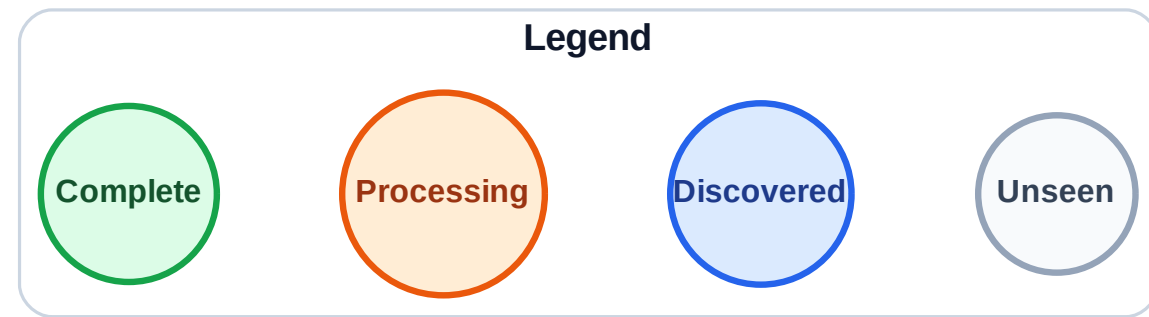
Queue: F → L → M → C → H → J → K

Visited: A, B



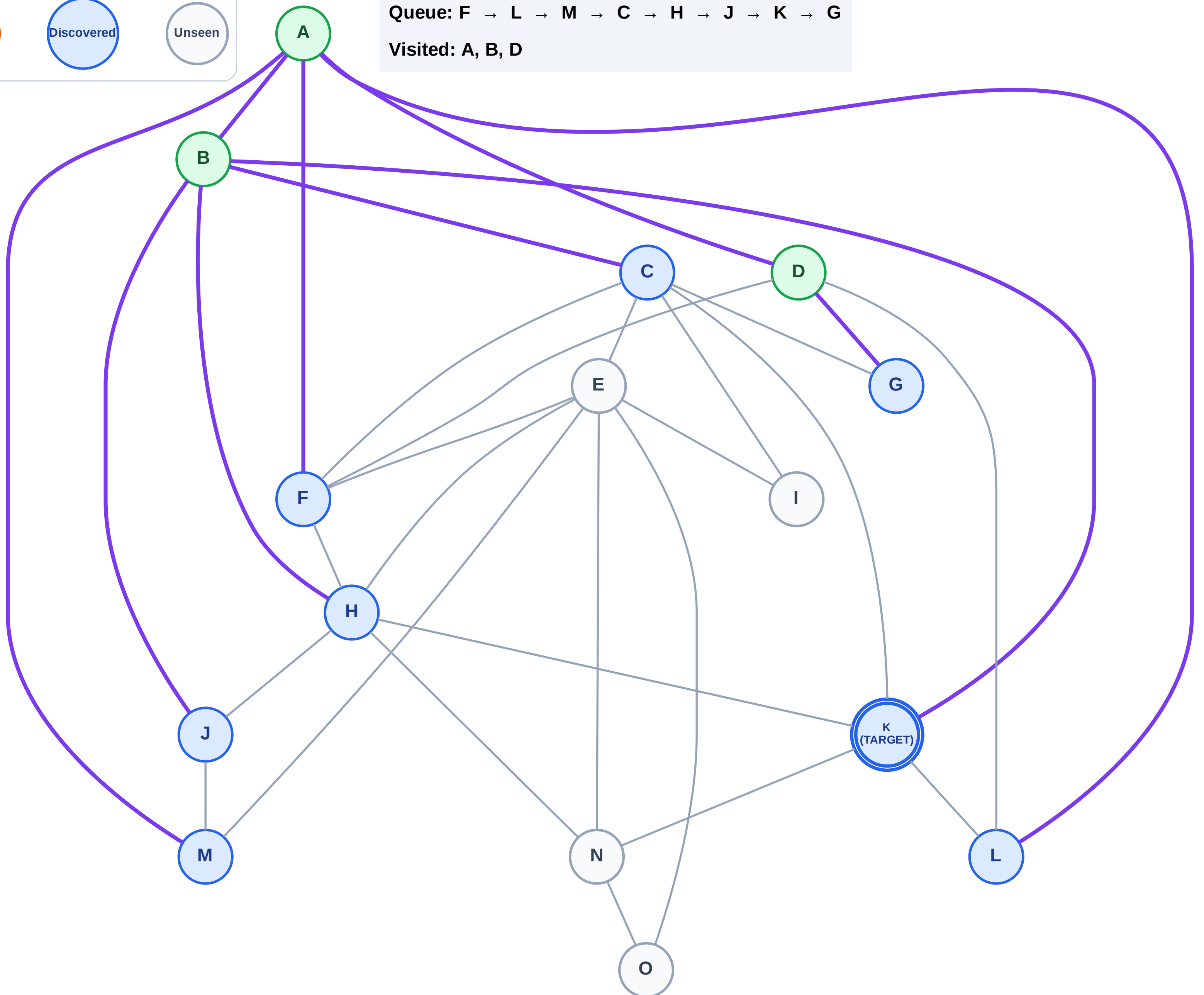
BFS Traversal

Step 7: Discover G from D



Queue: F → L → M → C → H → J → K → G

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

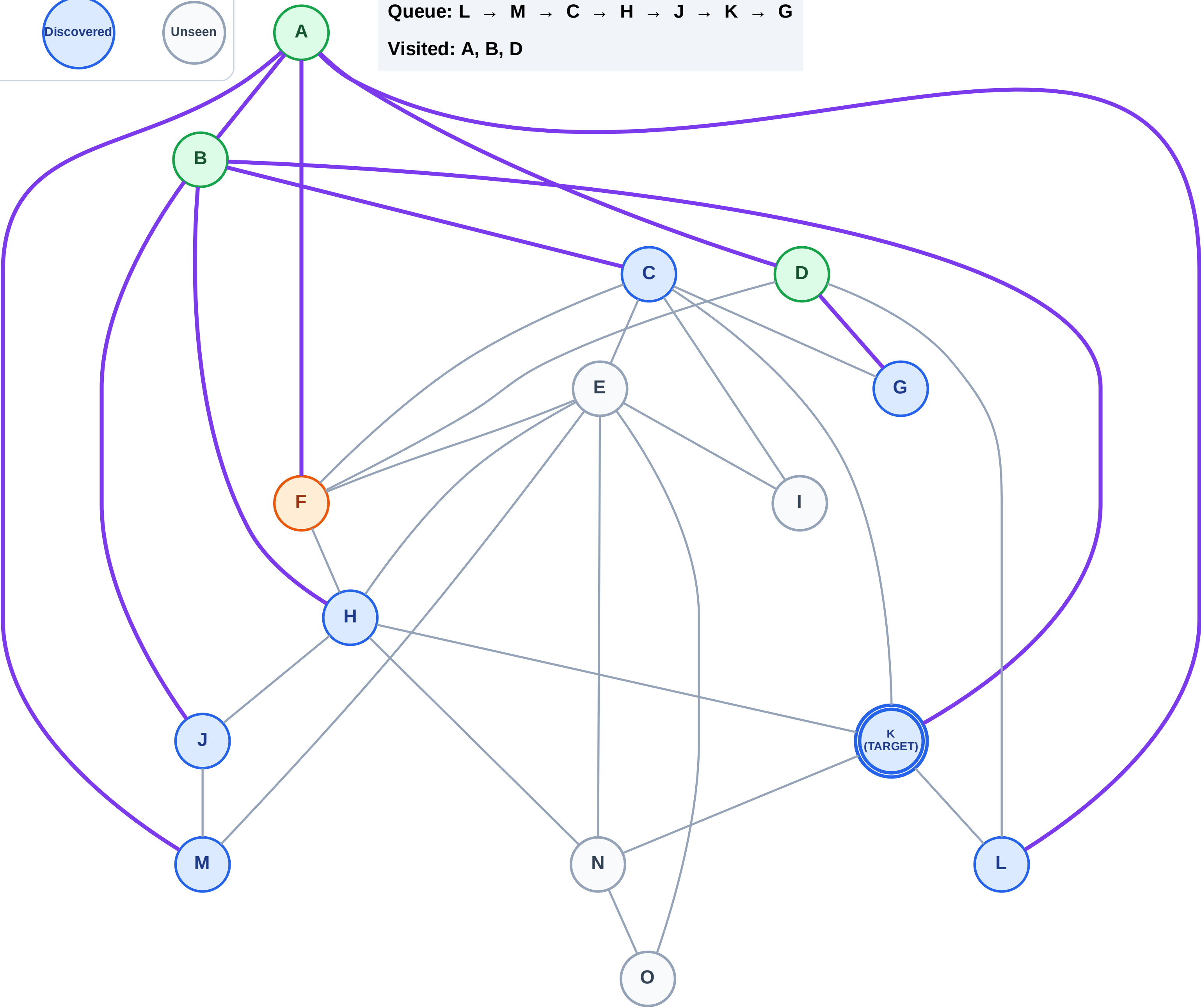
Processing

Discovered

Unseen

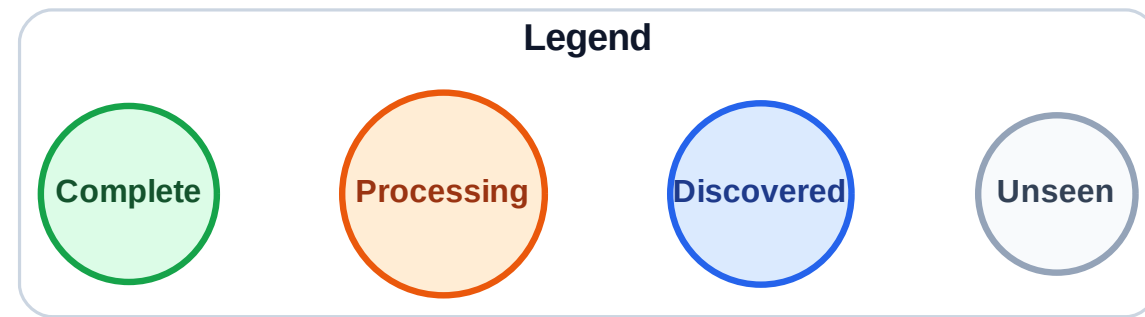
Queue: L → M → C → H → J → K → G

Visited: A, B, D



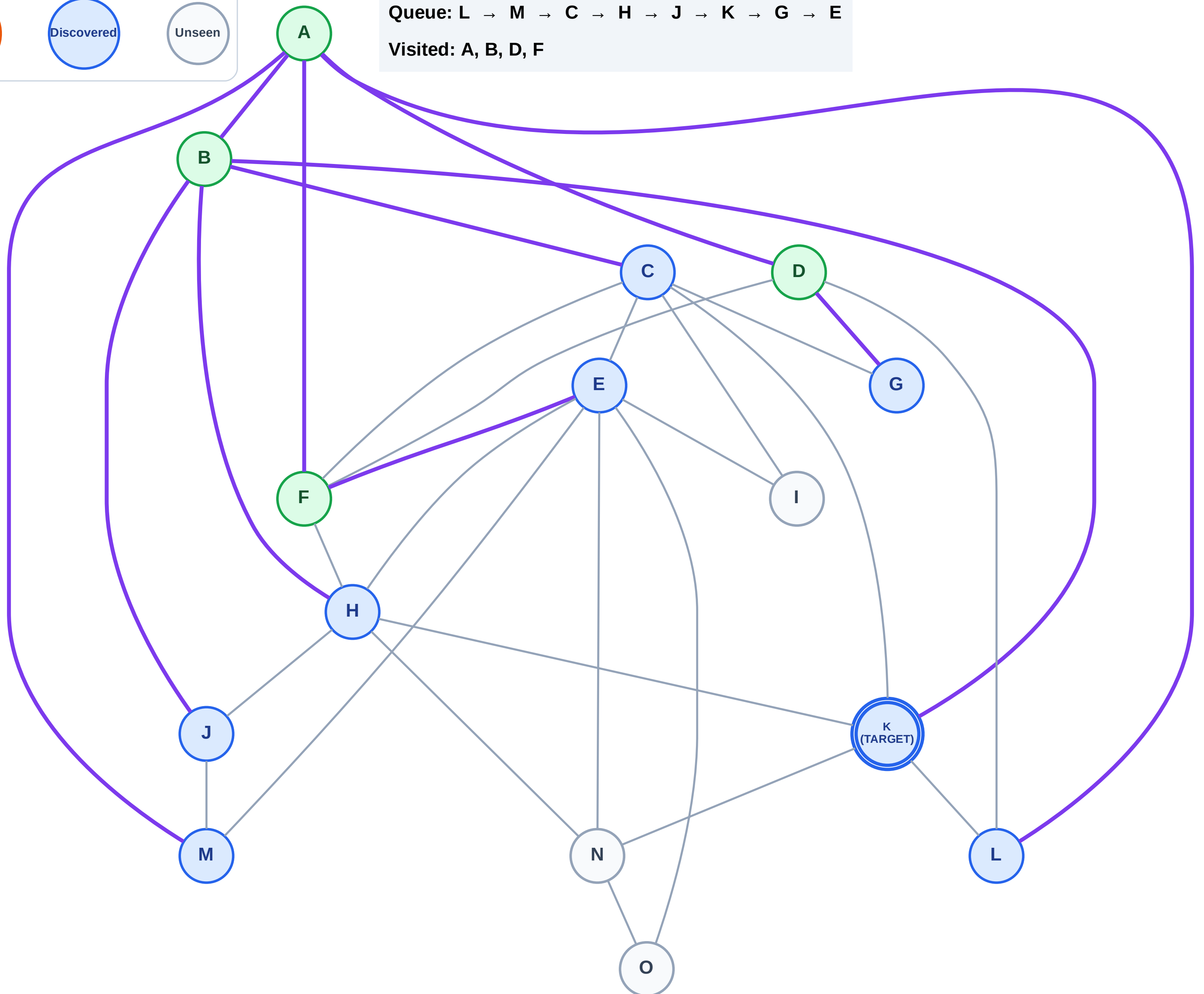
BFS Traversal

Step 9: Discover E from F



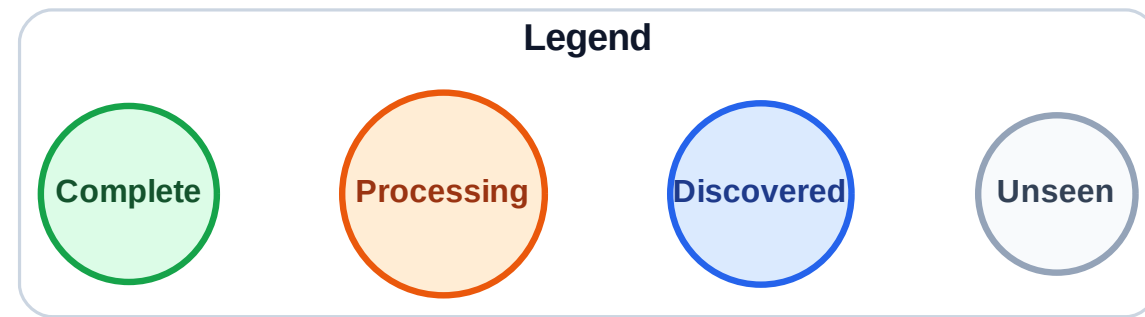
Queue: L → M → C → H → J → K → G → E

Visited: A, B, D, F



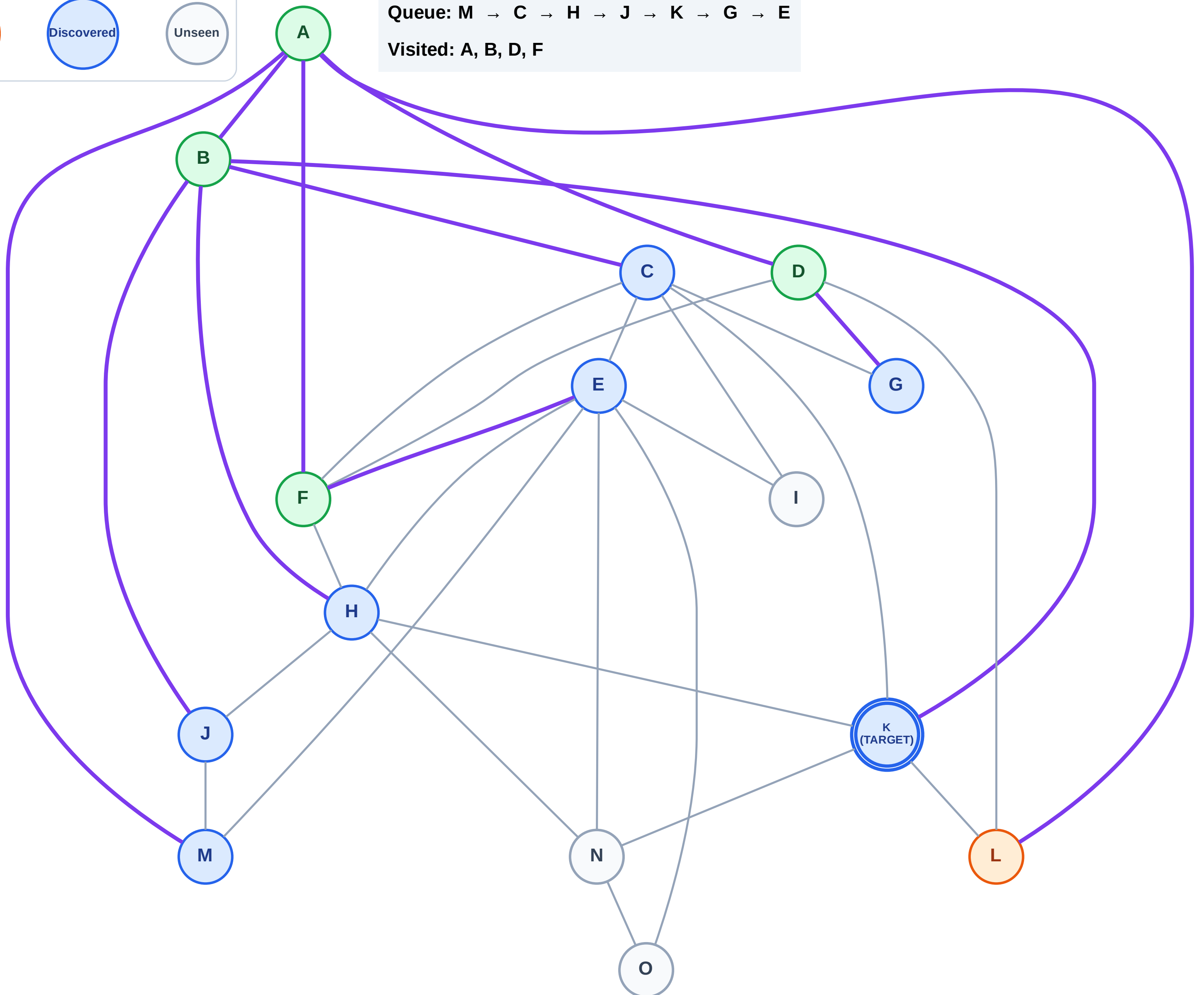
BFS Traversal

Step 10: Process node L



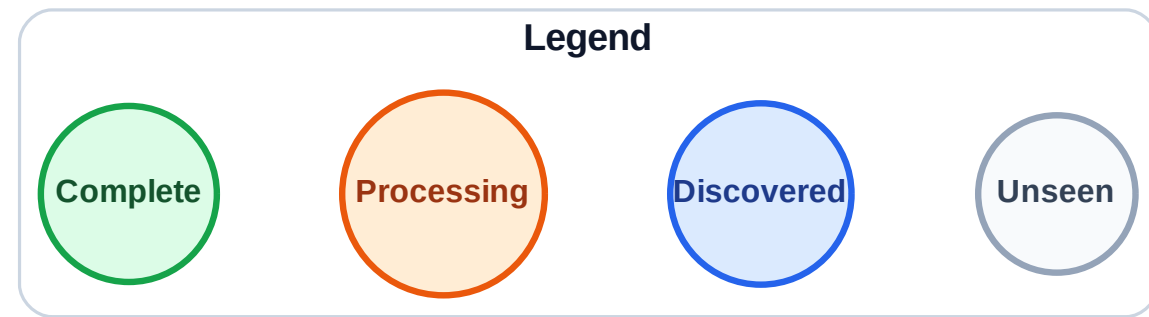
Queue: M → C → H → J → K → G → E

Visited: A, B, D, F

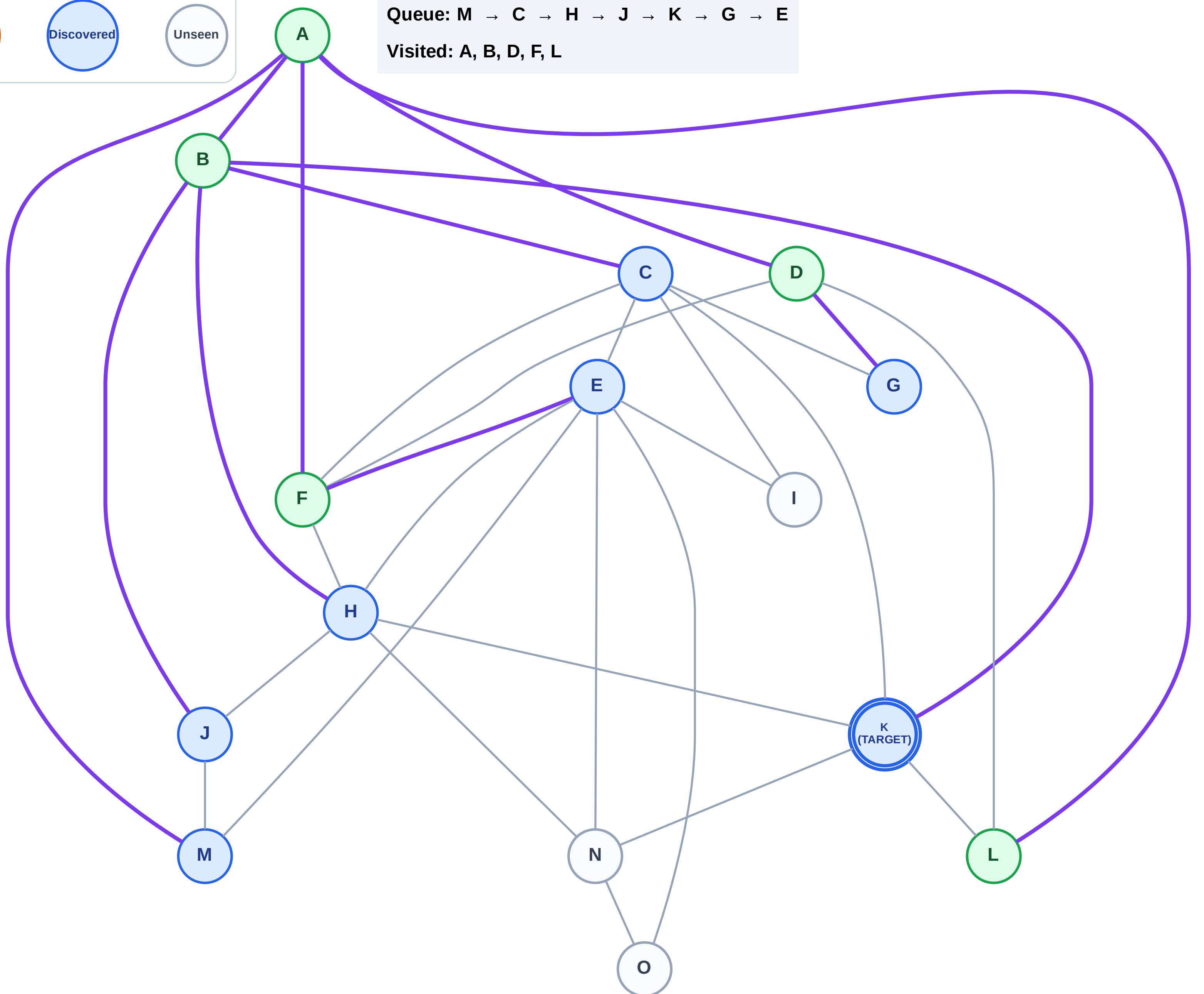


BFS Traversal

Step 11: No new neighbors from L

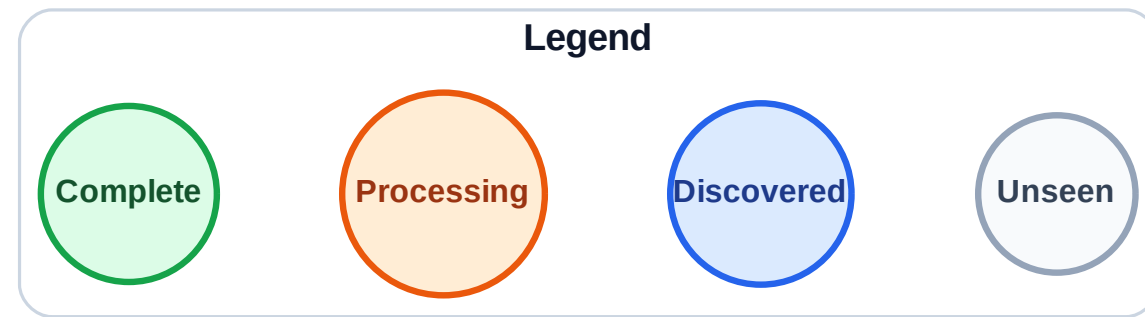


Queue: M → C → H → J → K → G → E
Visited: A, B, D, F, L



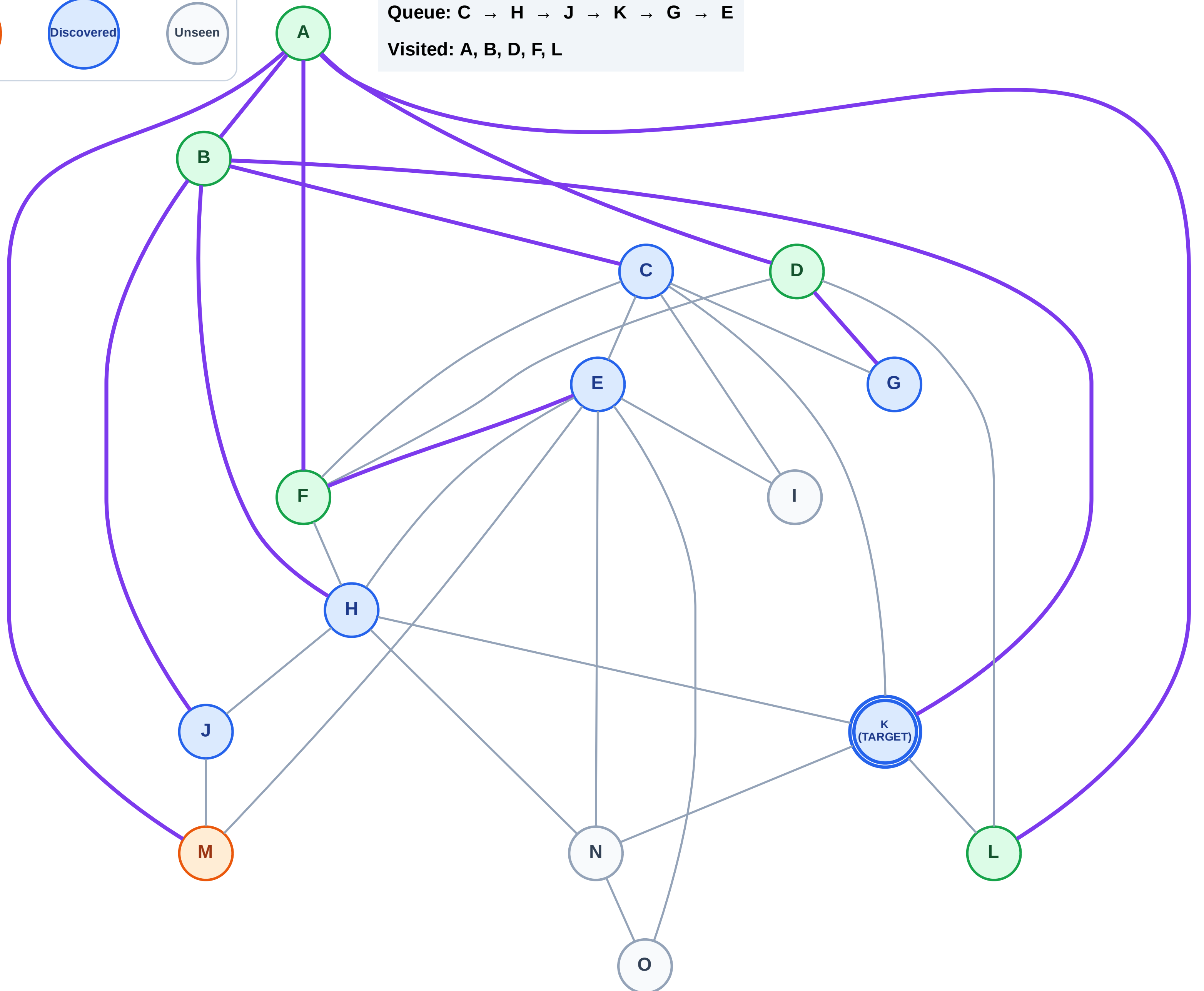
BFS Traversal

Step 12: Process node M



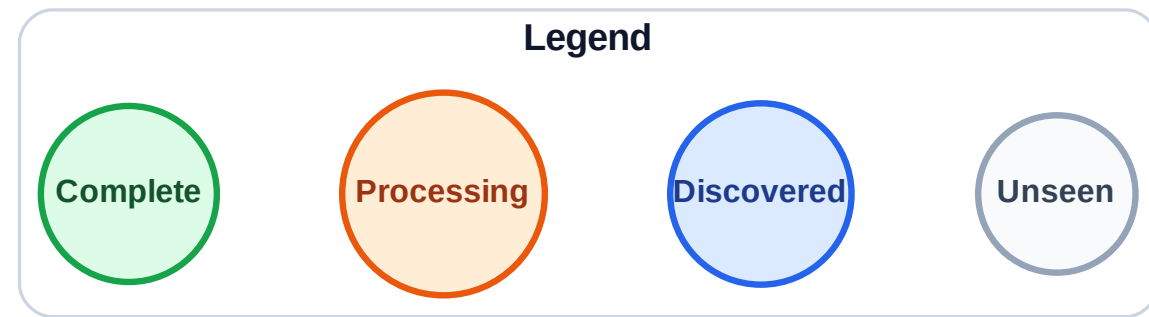
Queue: C → H → J → K → G → E

Visited: A, B, D, F, L



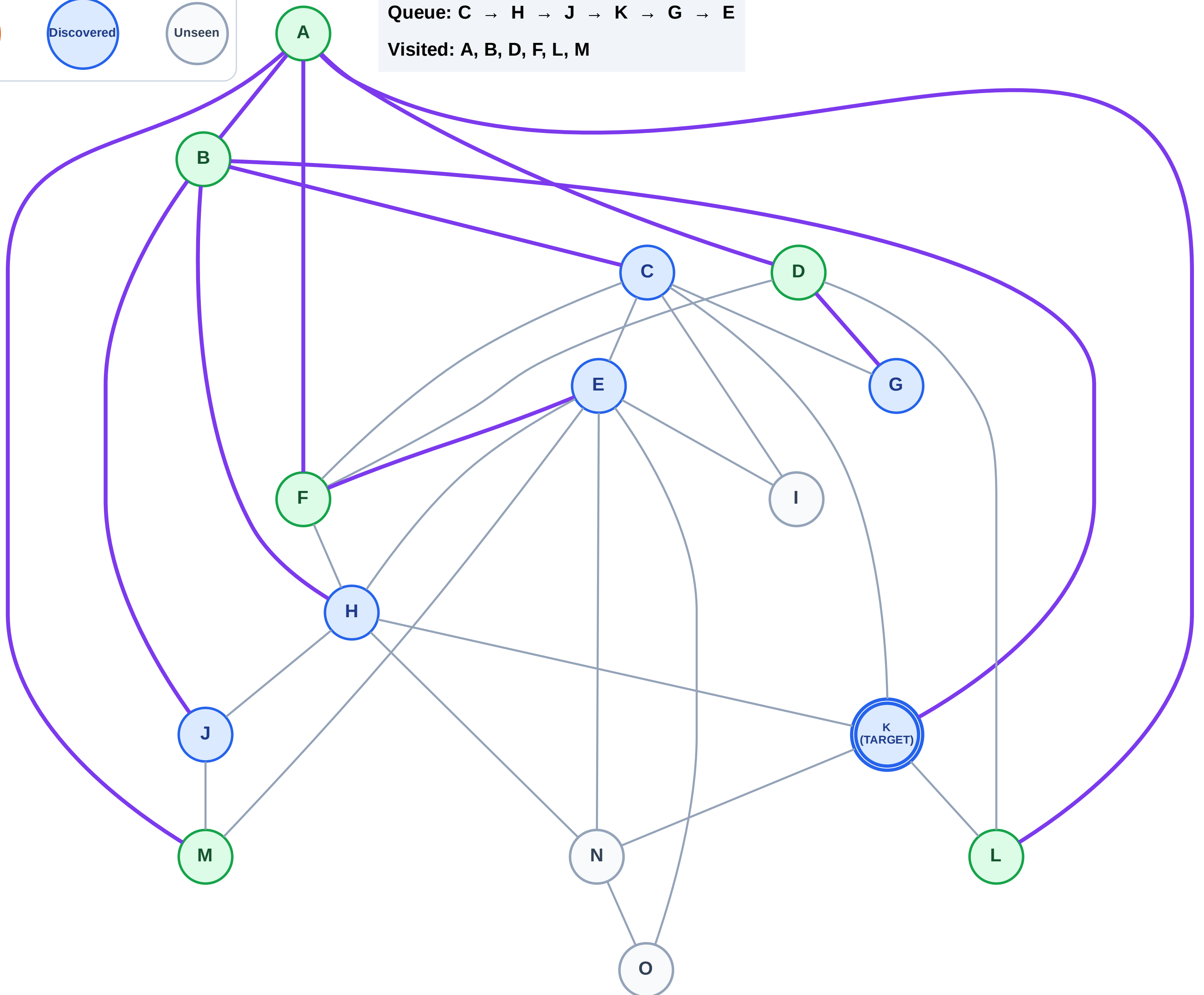
BFS Traversal

Step 13: No new neighbors from M



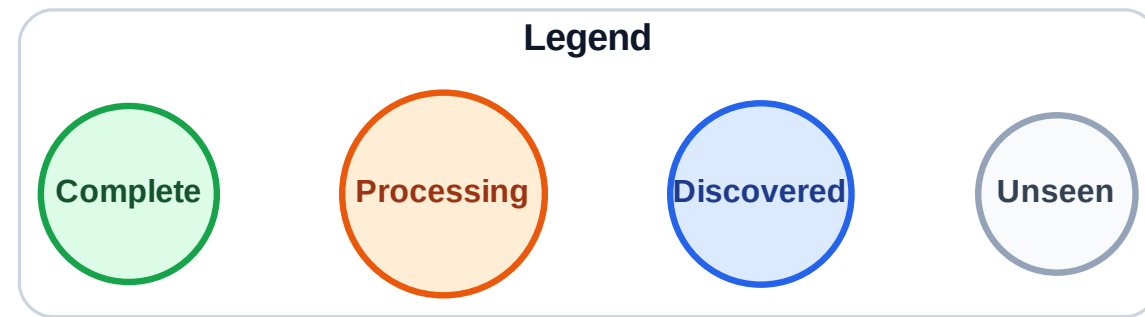
Queue: C → H → J → K → G → E

Visited: A, B, D, F, L, M



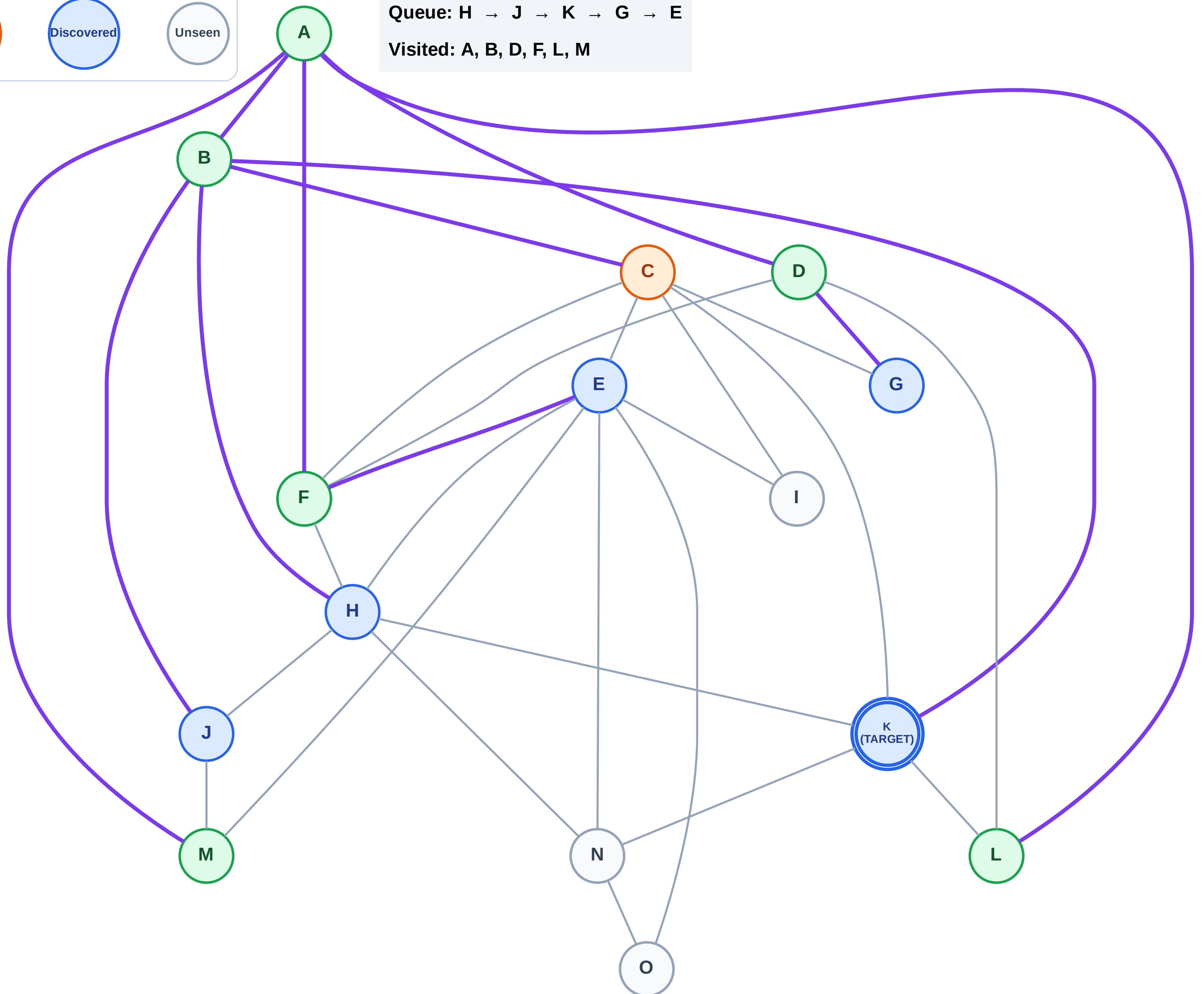
BFS Traversal

Step 14: Process node C



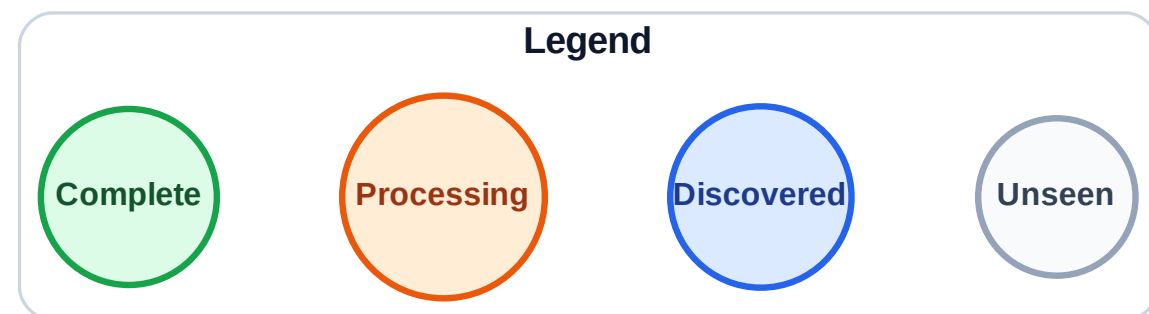
Queue: H → J → K → G → E

Visited: A, B, D, F, L, M



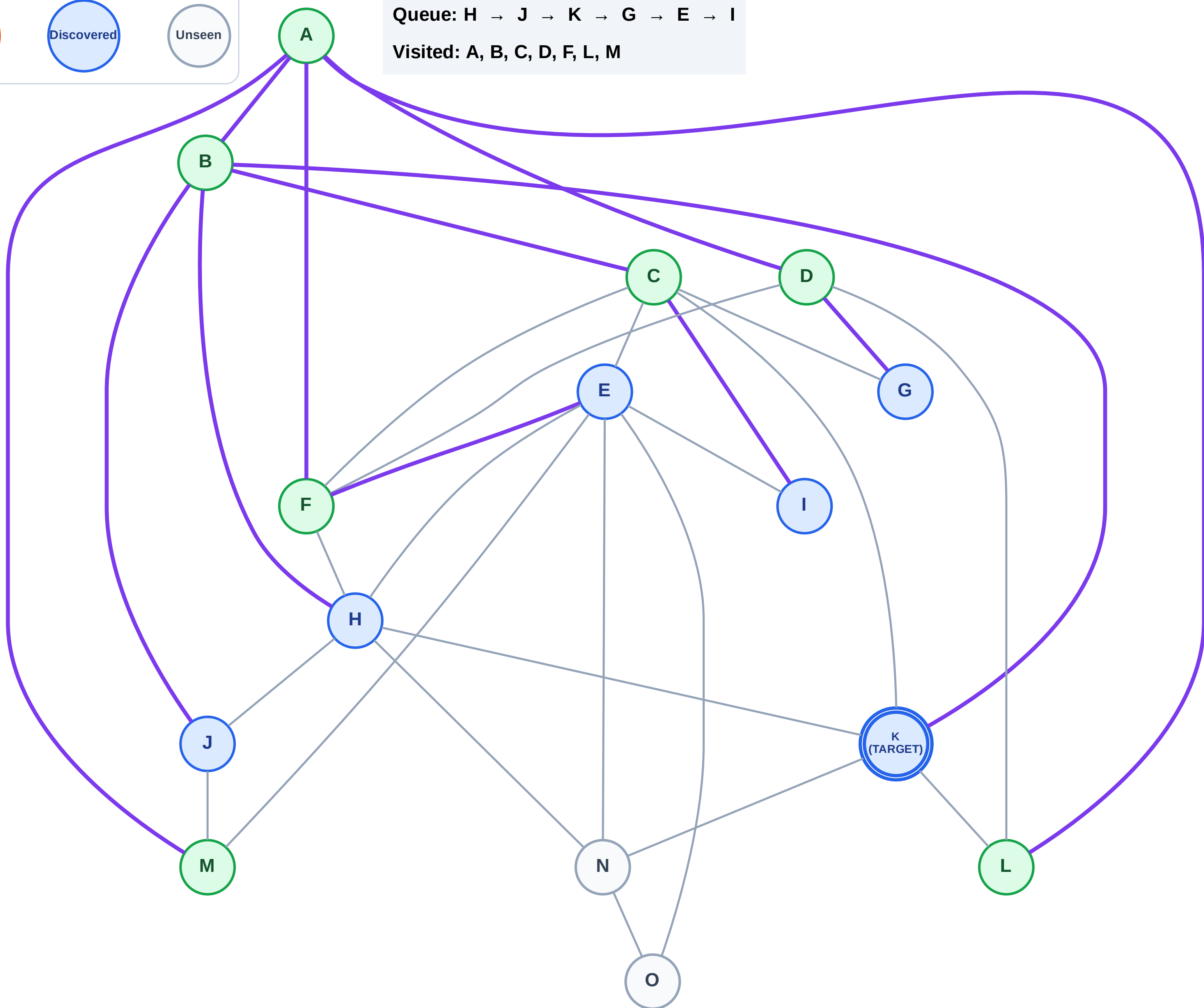
BFS Traversal

Step 15: Discover I from C



Queue: H → J → K → G → E → I

Visited: A, B, C, D, F, L, M



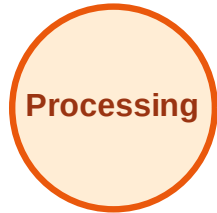
BFS Traversal

Step 16: Process node H

Legend



Complete



Processing



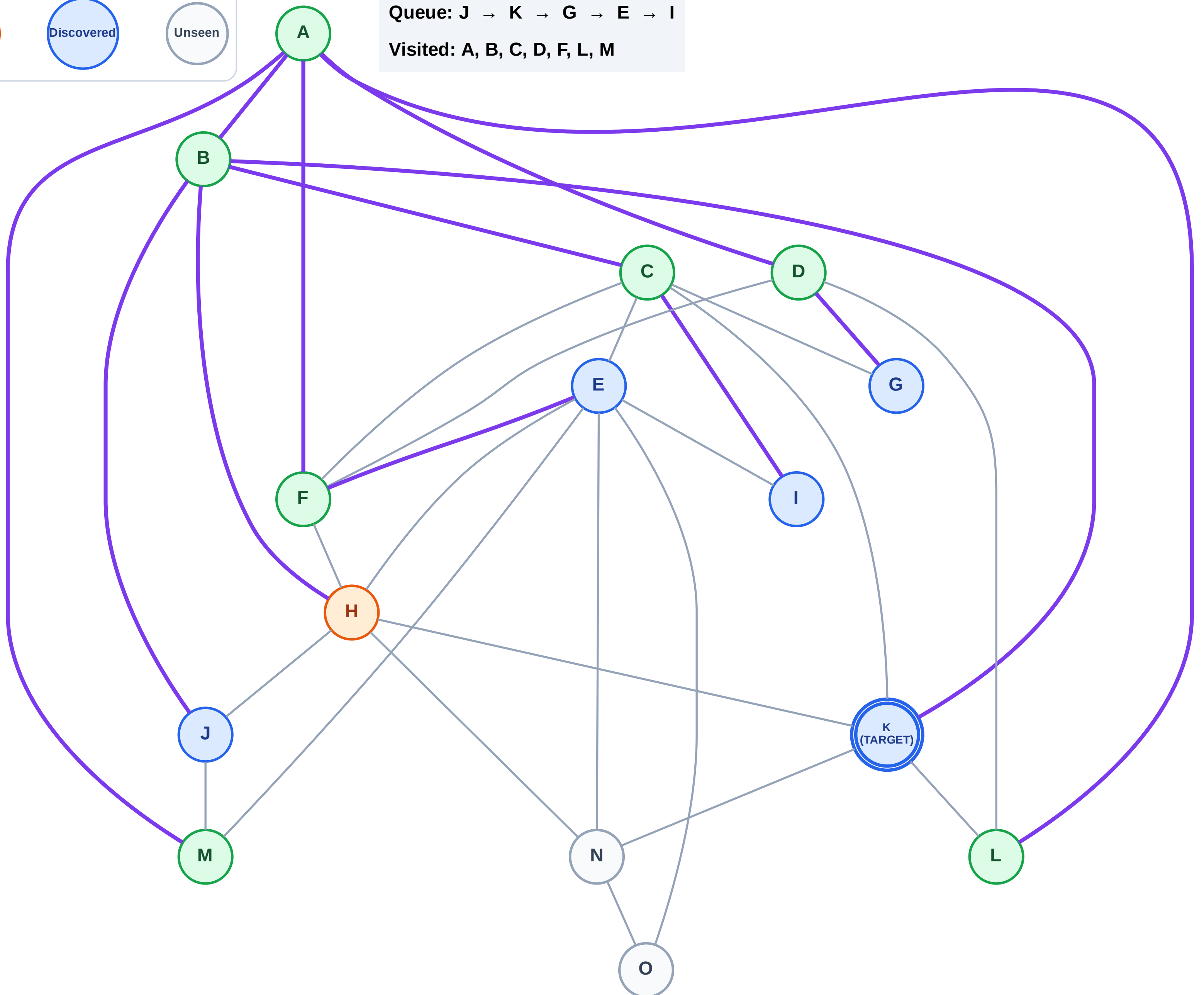
Discovered



Unseen

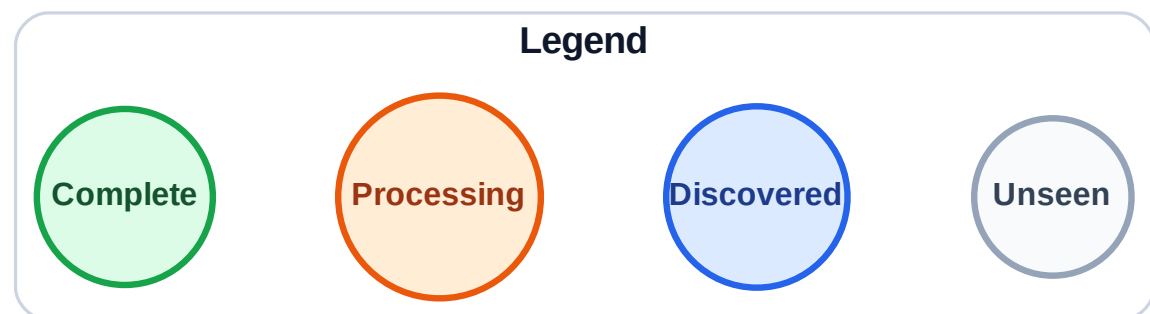
Queue: J → K → G → E → I

Visited: A, B, C, D, F, L, M



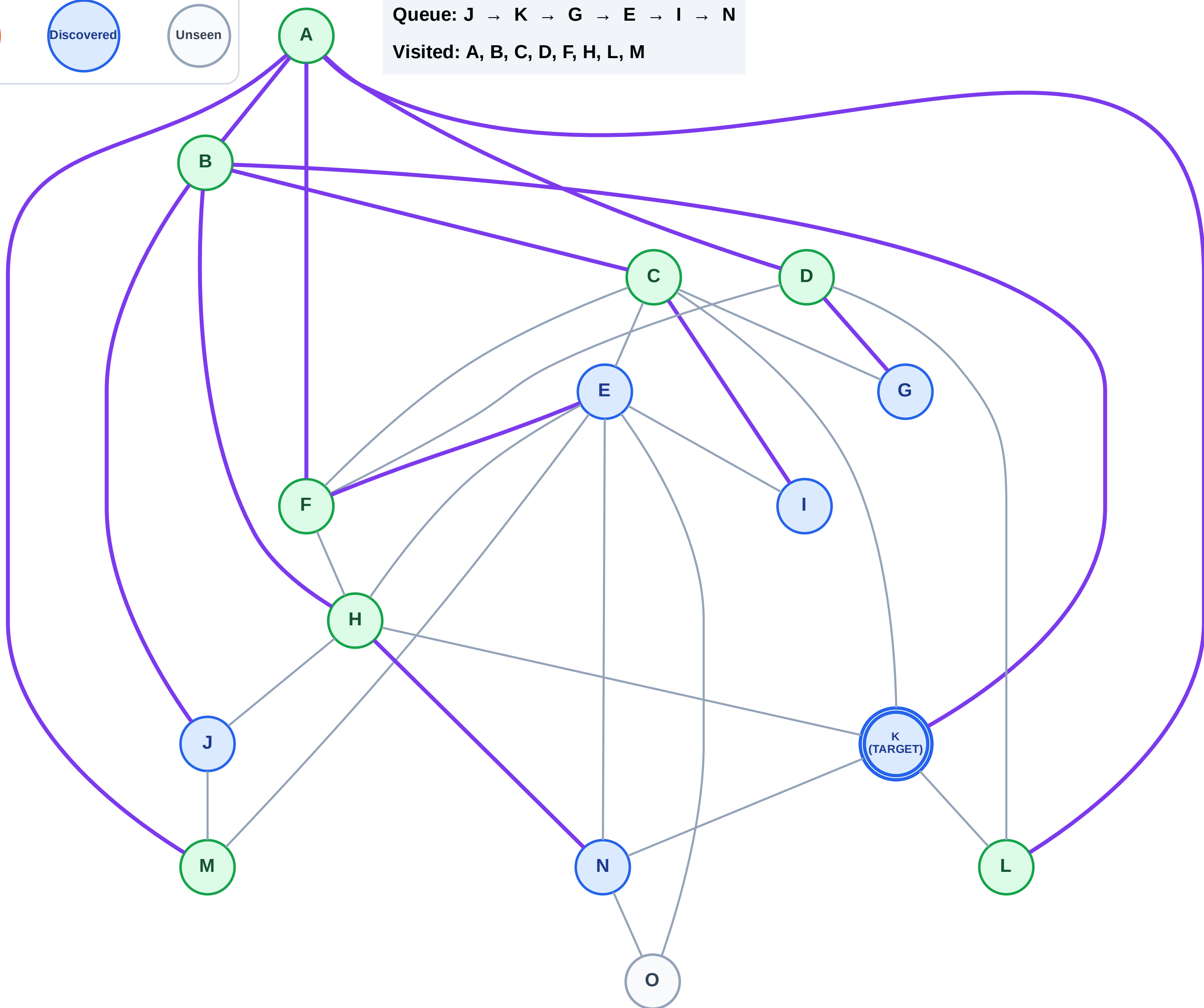
BFS Traversal

Step 17: Discover N from H



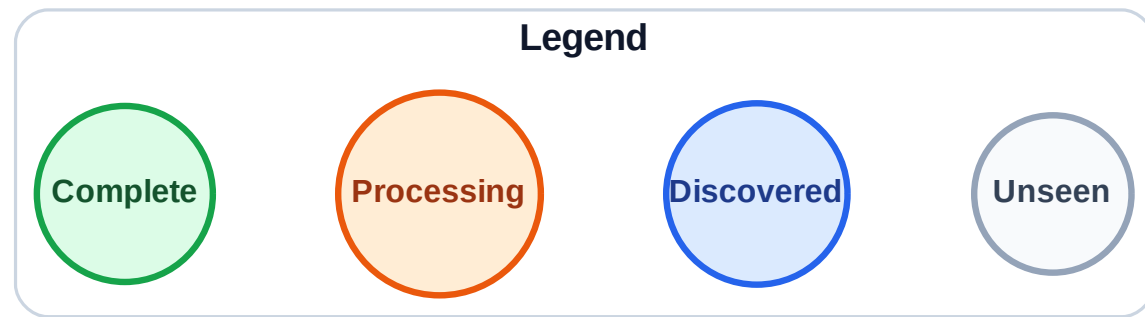
Queue: J → K → G → E → I → N

Visited: A, B, C, D, F, H, L, M



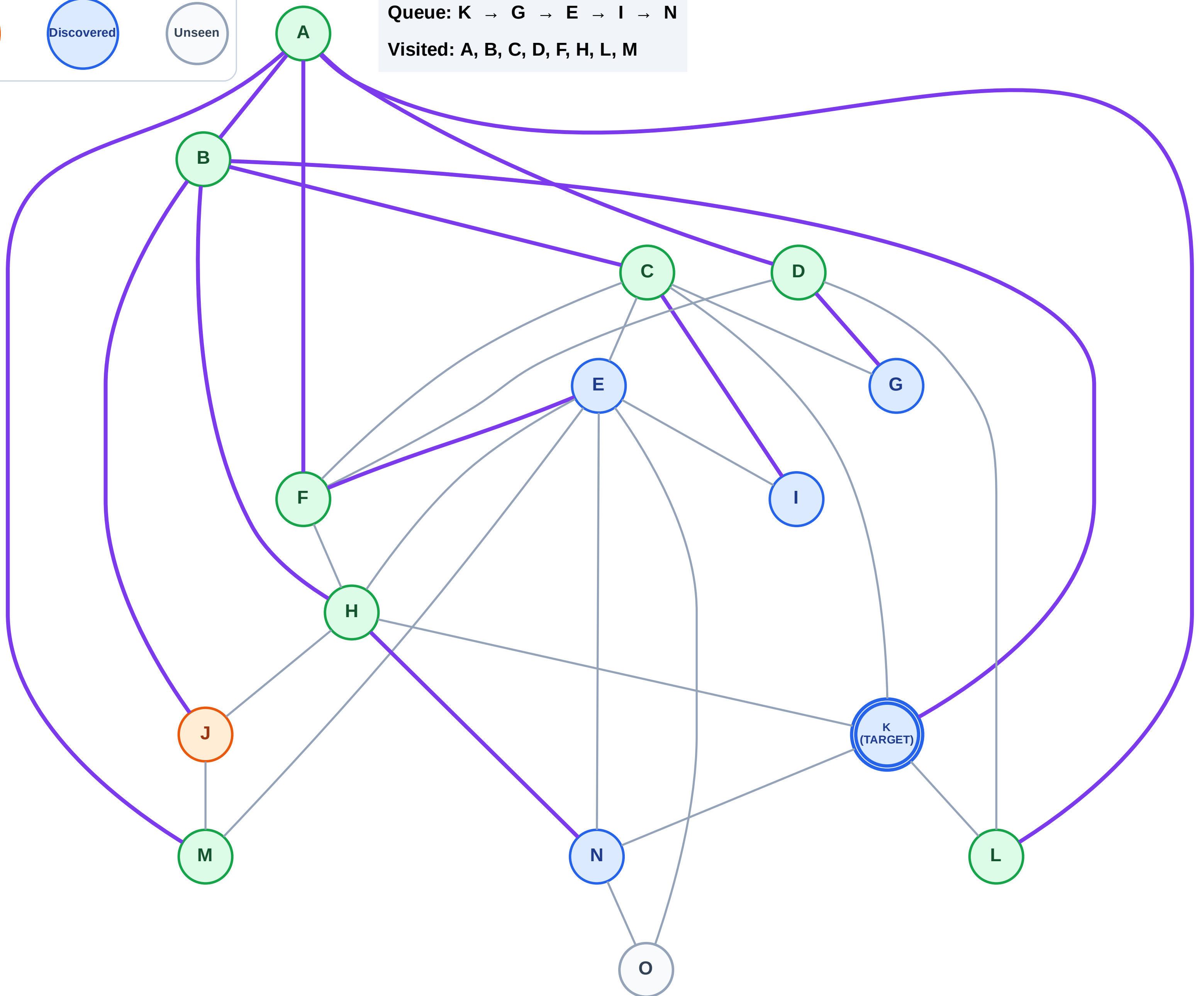
BFS Traversal

Step 18: Process node J



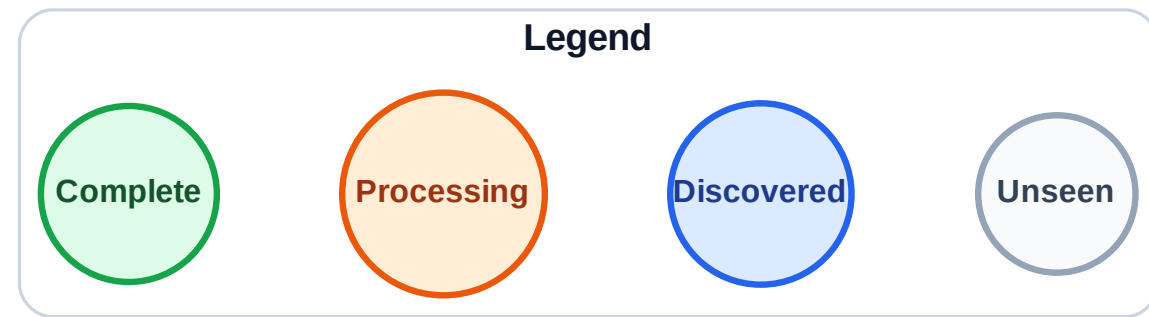
Queue: K → G → E → I → N

Visited: A, B, C, D, F, H, L, M



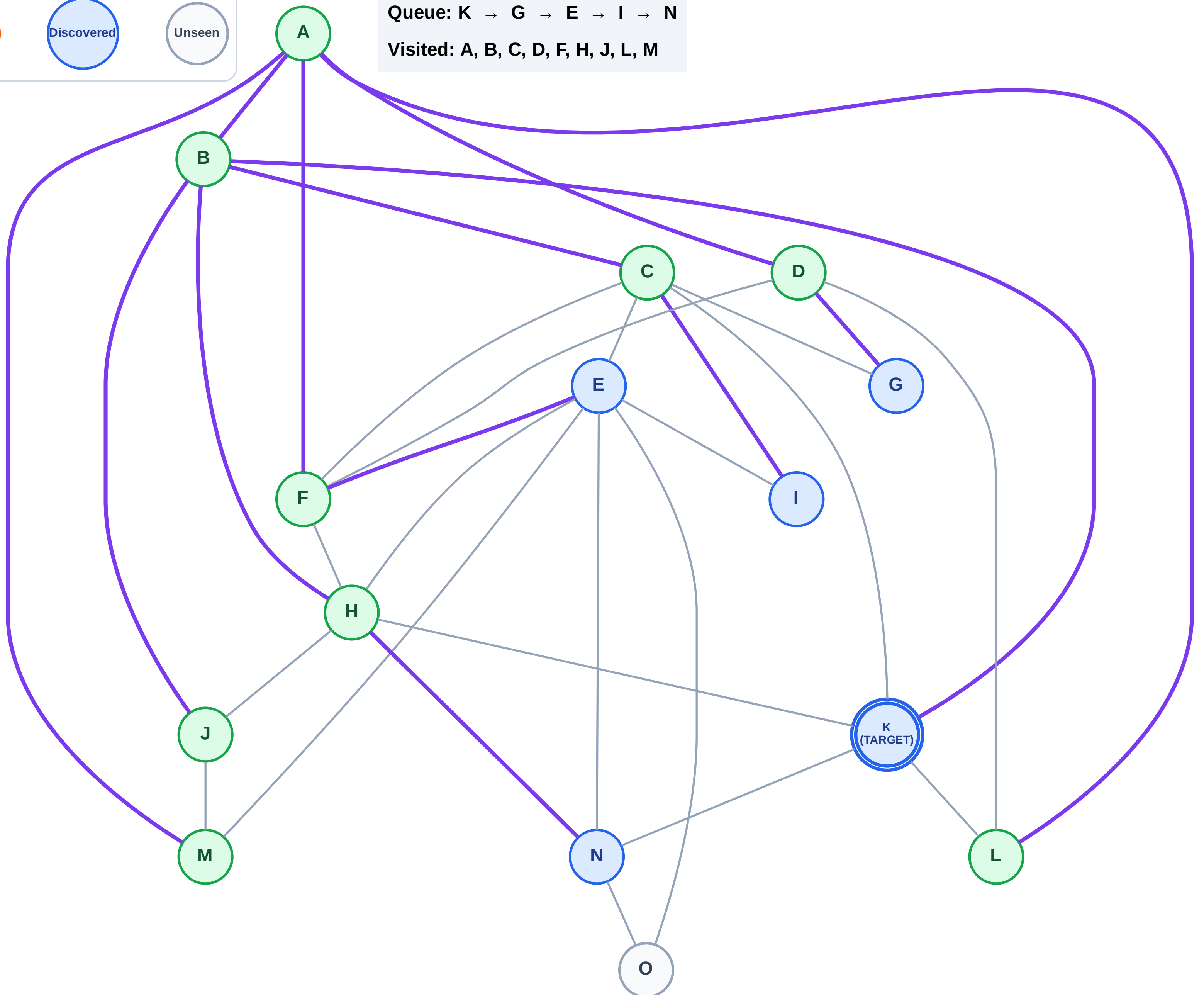
BFS Traversal

Step 19: No new neighbors from J



Queue: K → G → E → I → N

Visited: A, B, C, D, F, H, J, L, M



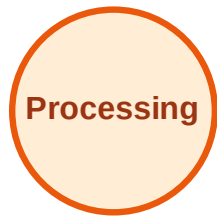
BFS Traversal

Step 20: Process node K

Legend



Complete



Processing



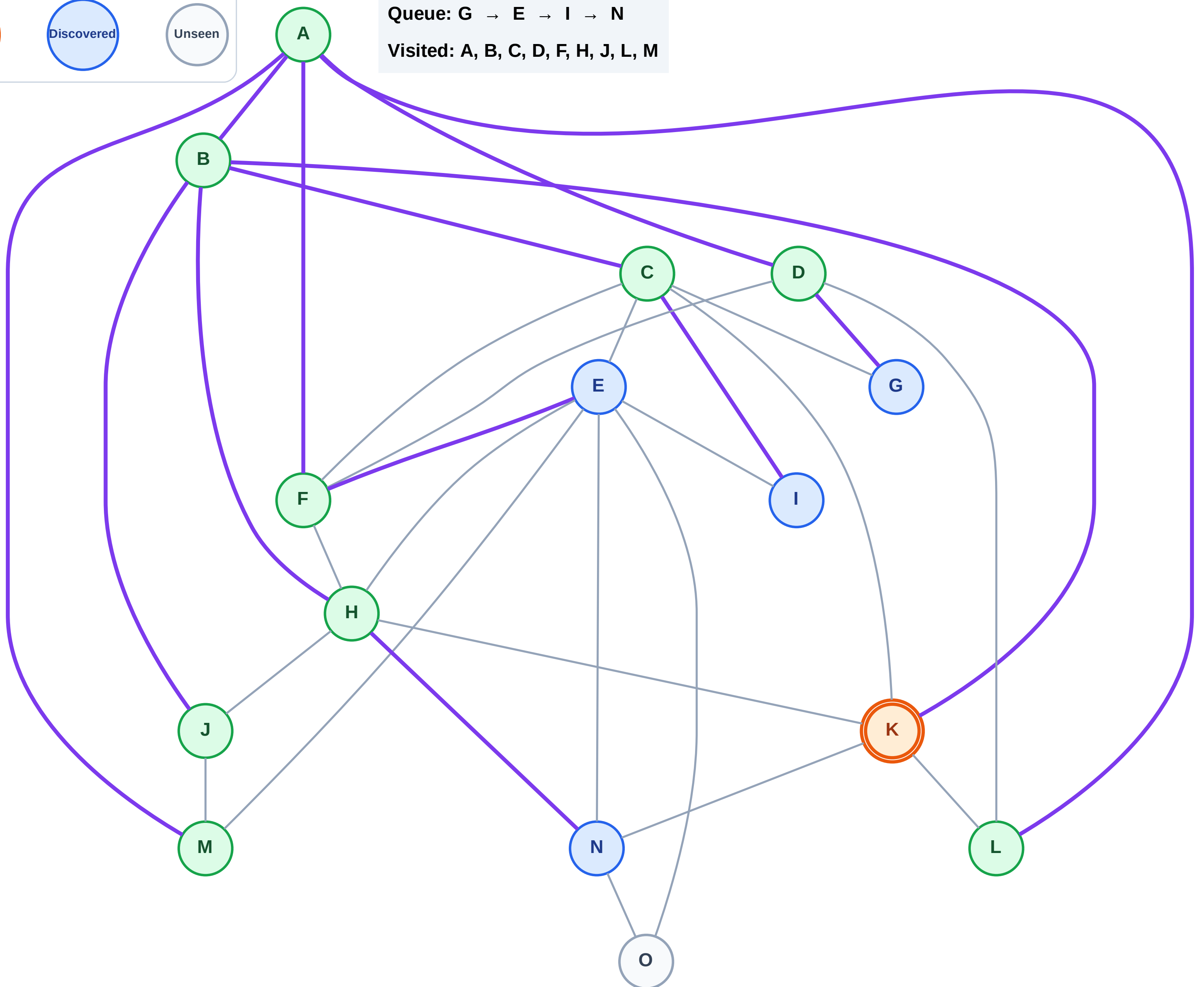
Discovered



Unseen

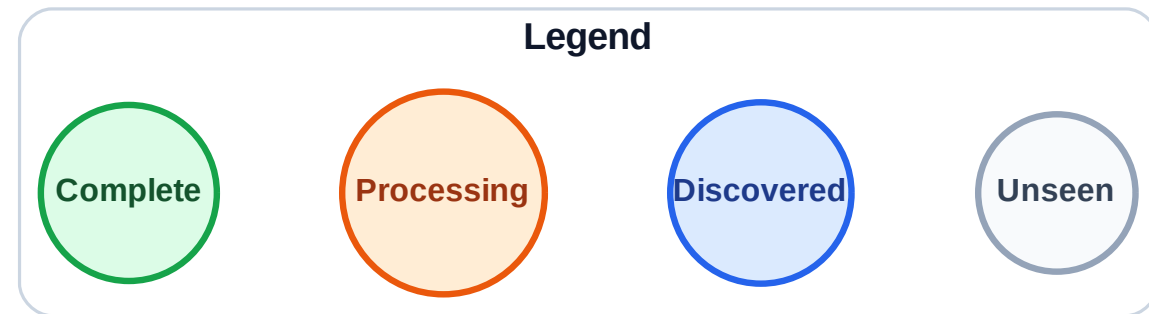
Queue: G → E → I → N

Visited: A, B, C, D, F, H, J, L, M



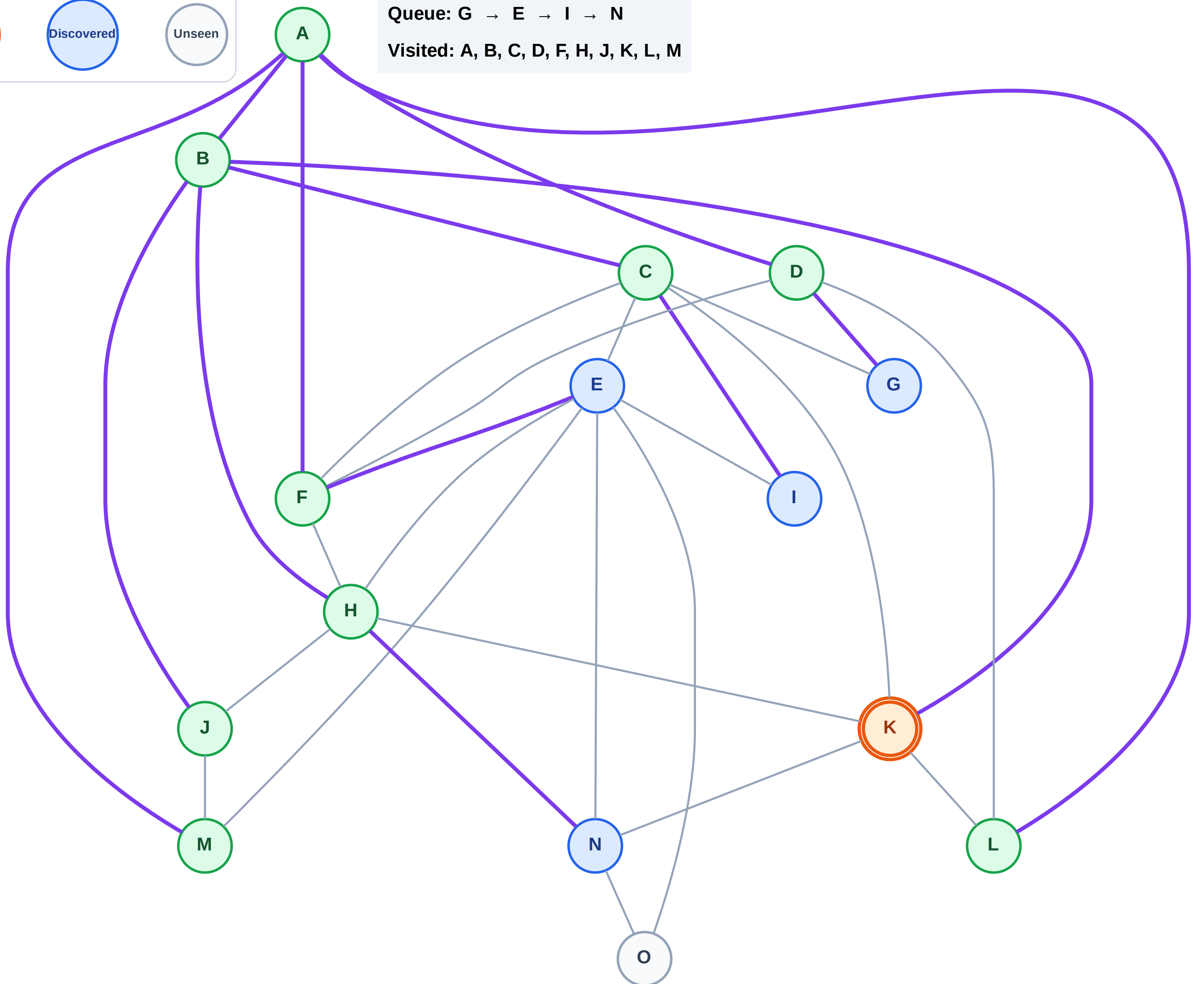
BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K



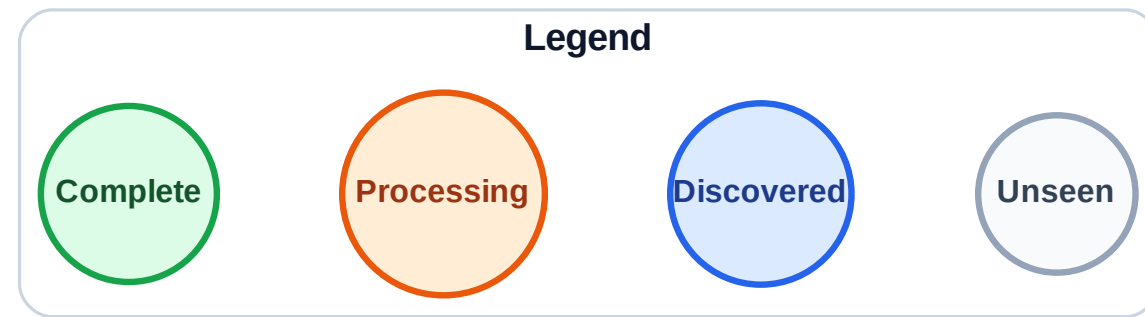
Queue: G → E → I → N

Visited: A, B, C, D, F, H, J, K, L, M

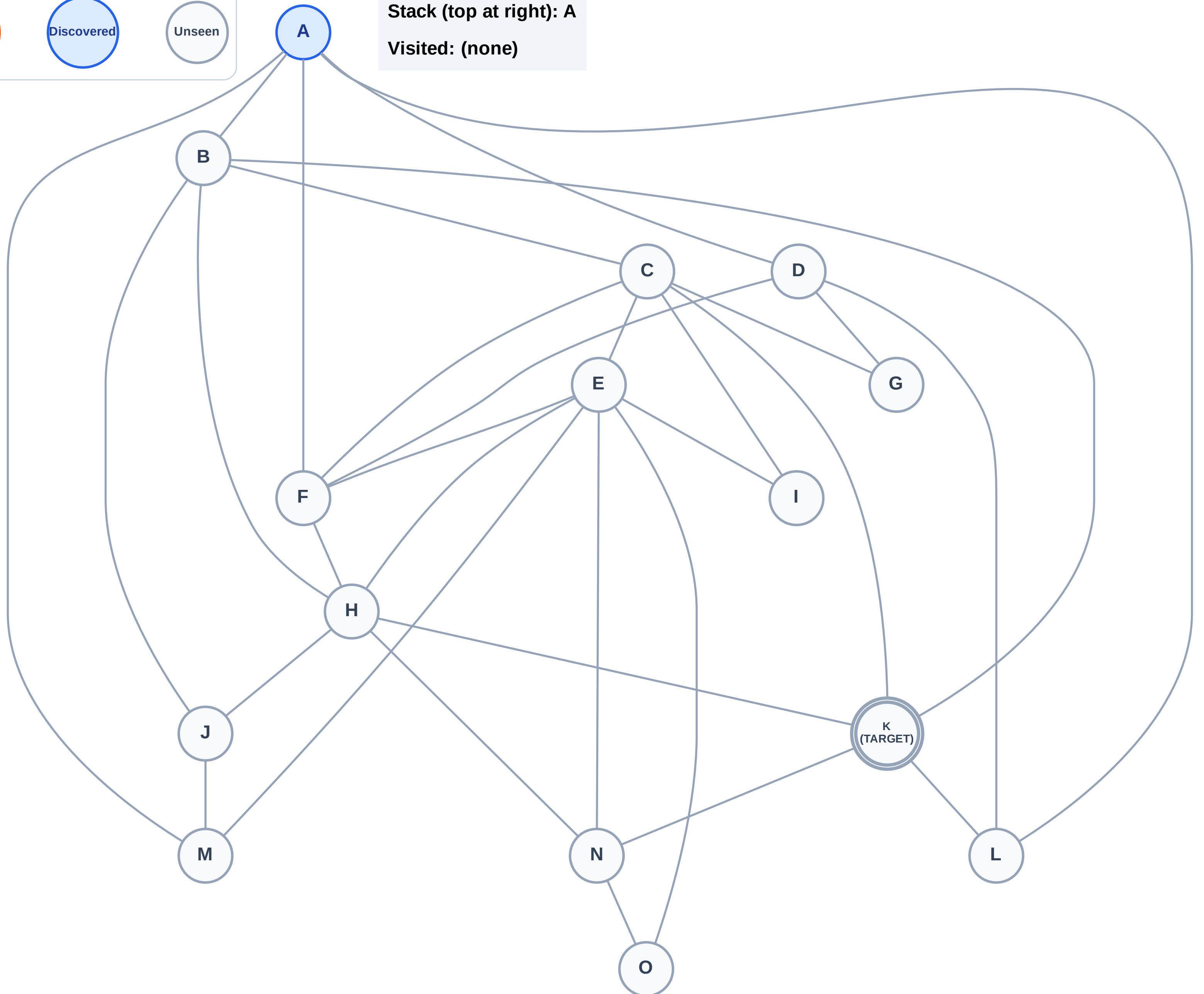


DFS Traversal

Step 1: Start at A (Seeking target node K)

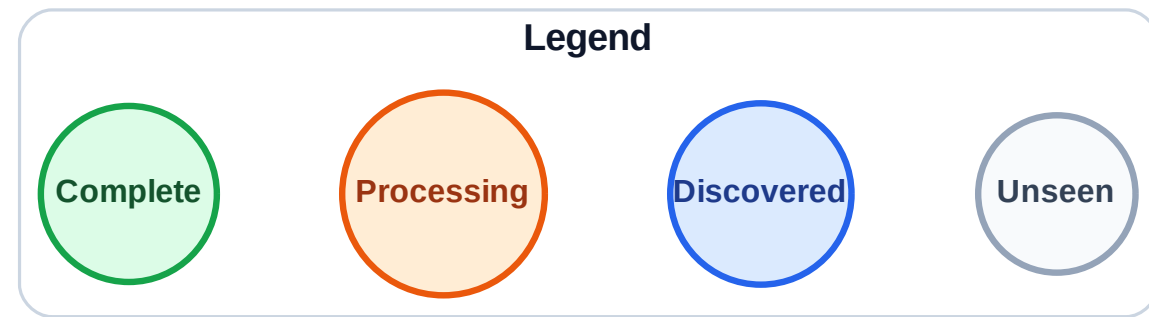


Stack (top at right): A
Visited: (none)



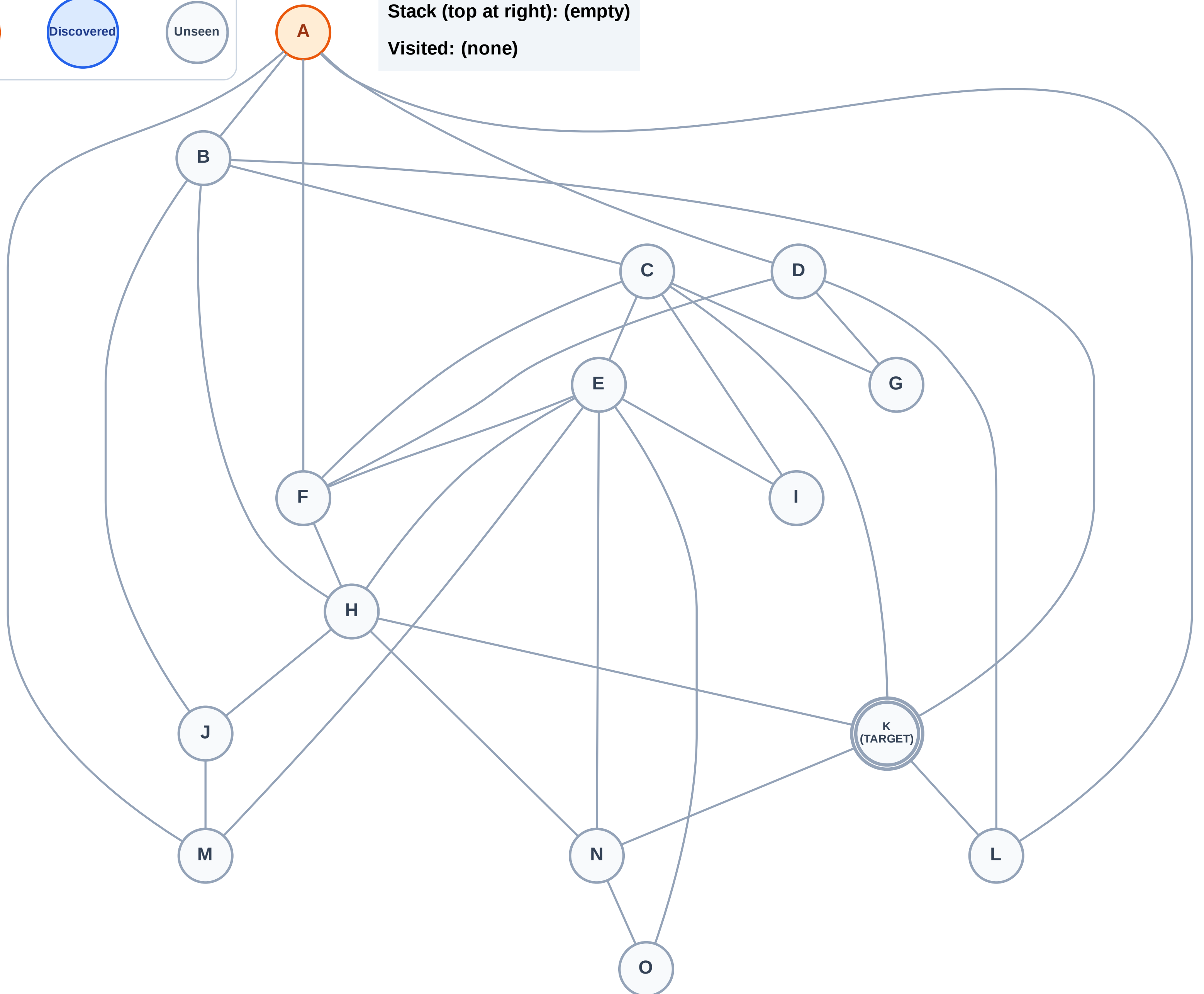
DFS Traversal

Step 2: Process node A



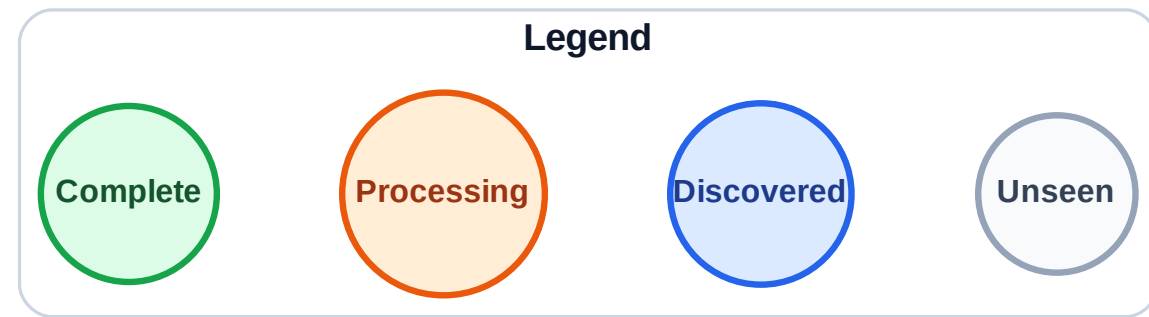
Stack (top at right): (empty)

Visited: (none)

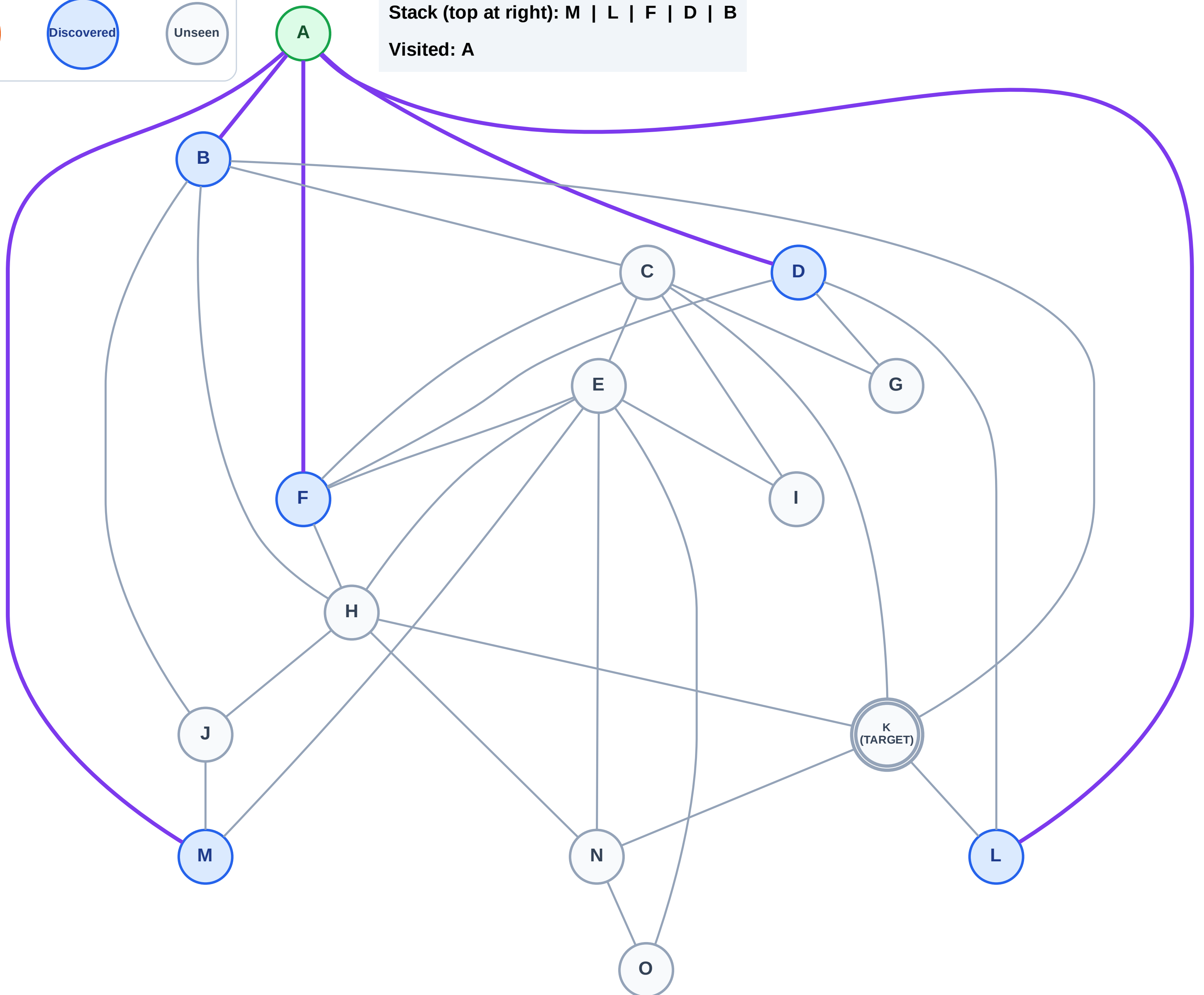


DFS Traversal

Step 3: Discover M, L, F, D, B from A



Stack (top at right): M | L | F | D | B
Visited: A



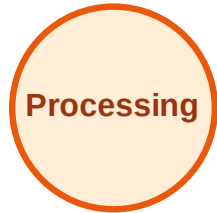
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



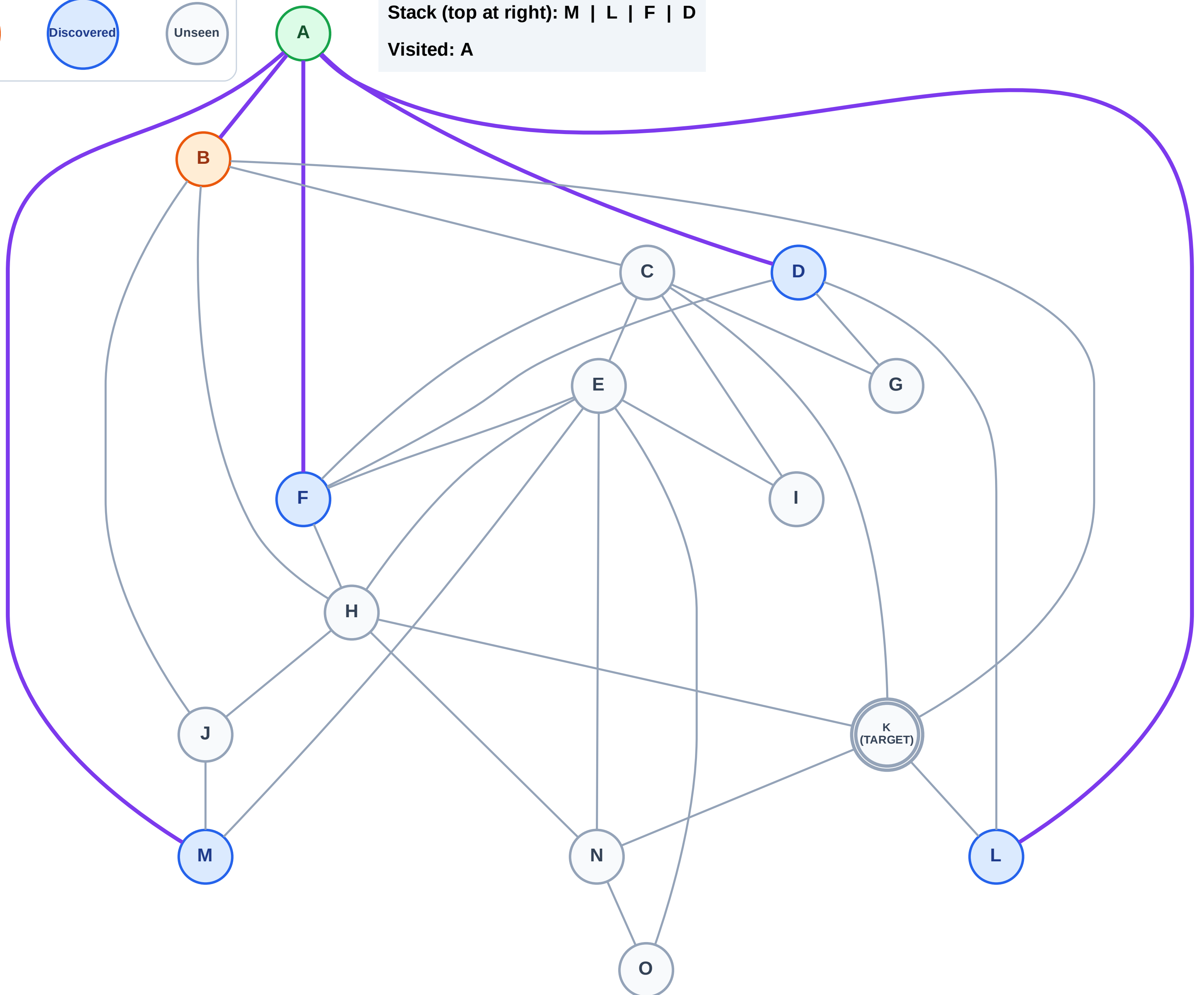
Discovered



Unseen

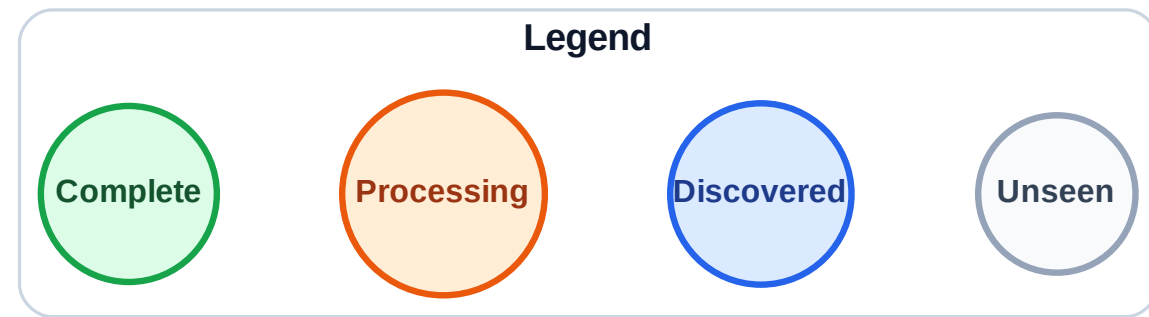
Stack (top at right): M | L | F | D

Visited: A

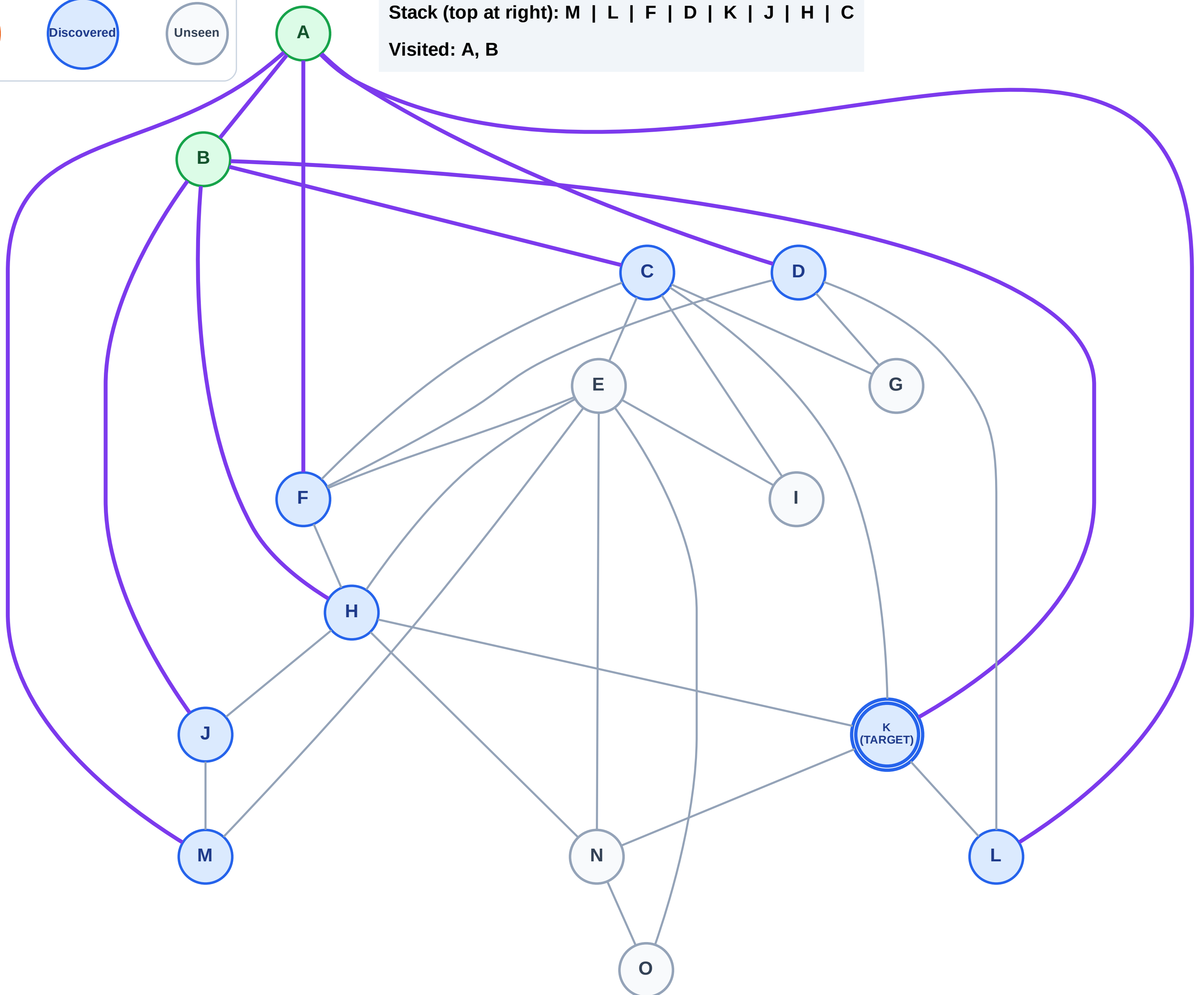


DFS Traversal

Step 5: Discover K, J, H, C from B

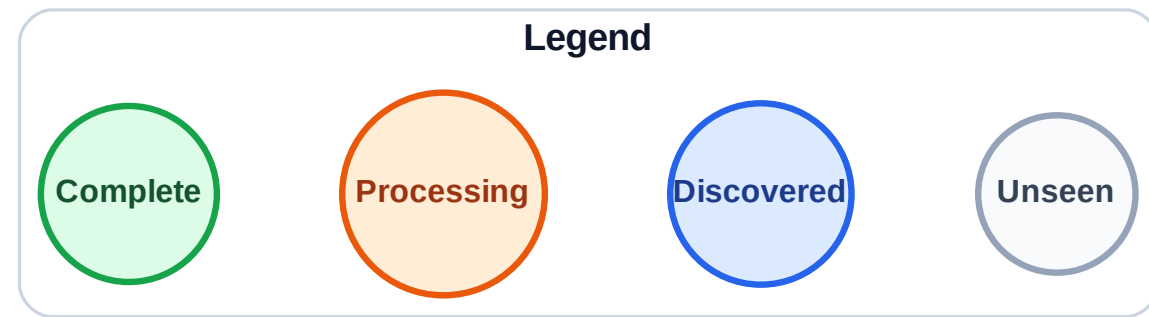


Stack (top at right): M | L | F | D | K | J | H | C
Visited: A, B



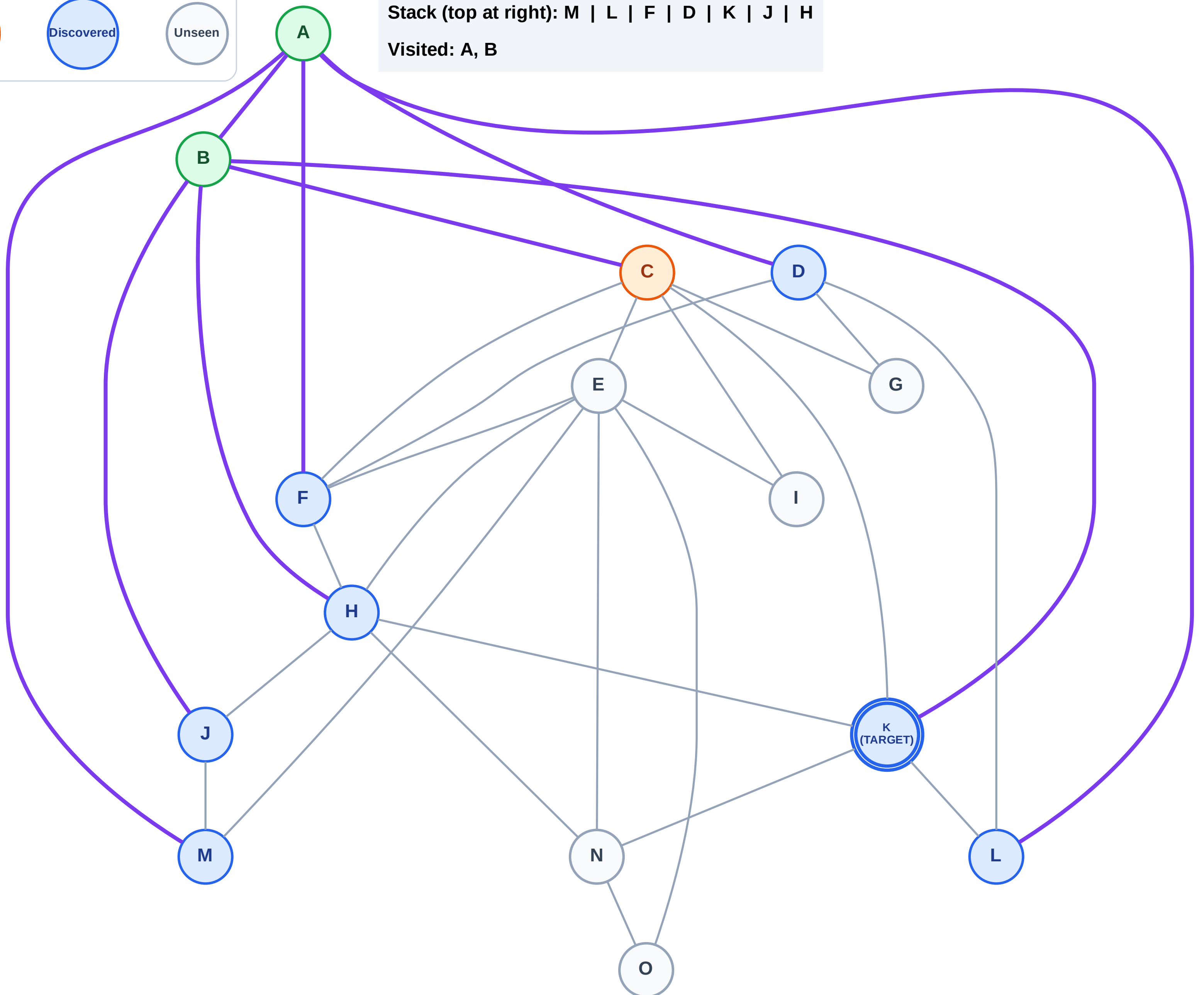
DFS Traversal

Step 6: Process node C



Stack (top at right): M | L | F | D | K | J | H

Visited: A, B



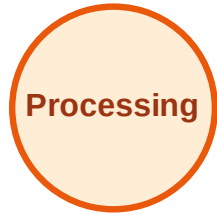
DFS Traversal

Step 7: Discover I, G, E from C

Legend



Complete



Processing



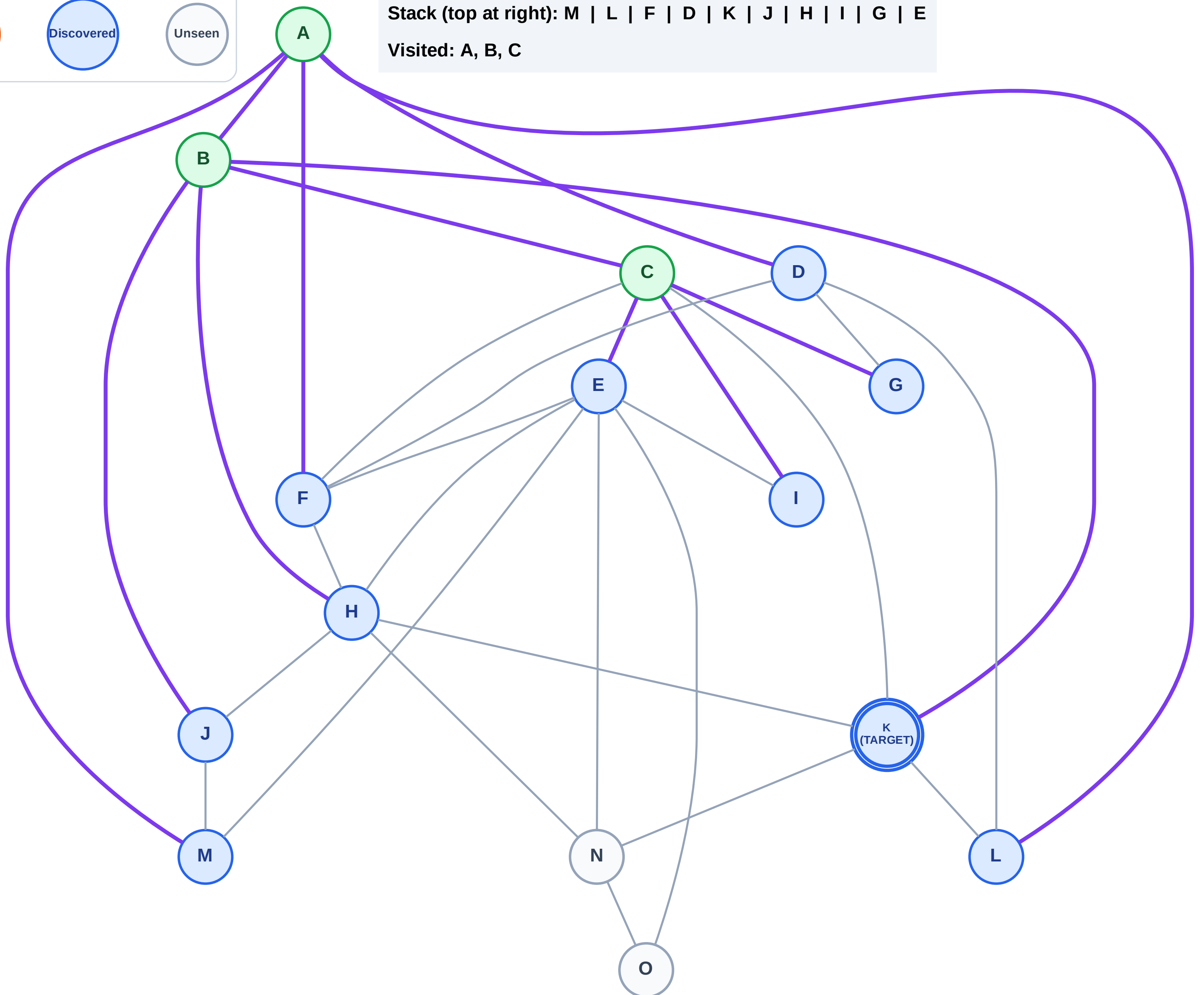
Discovered



Unseen

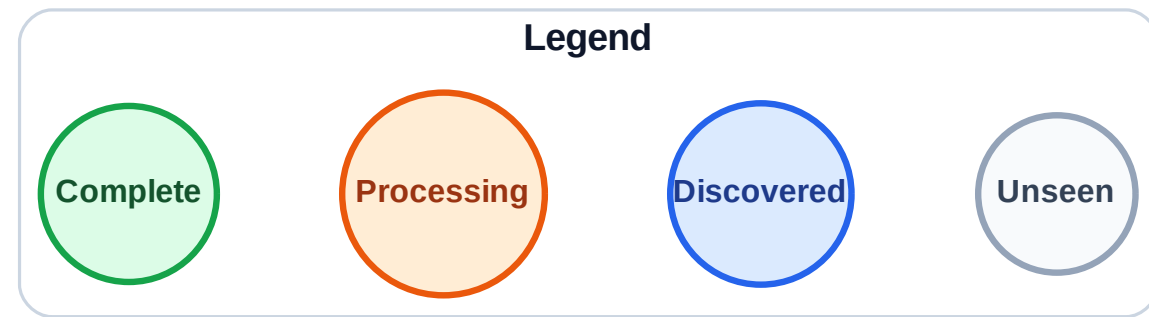
Stack (top at right): M | L | F | D | K | J | H | I | G | E

Visited: A, B, C

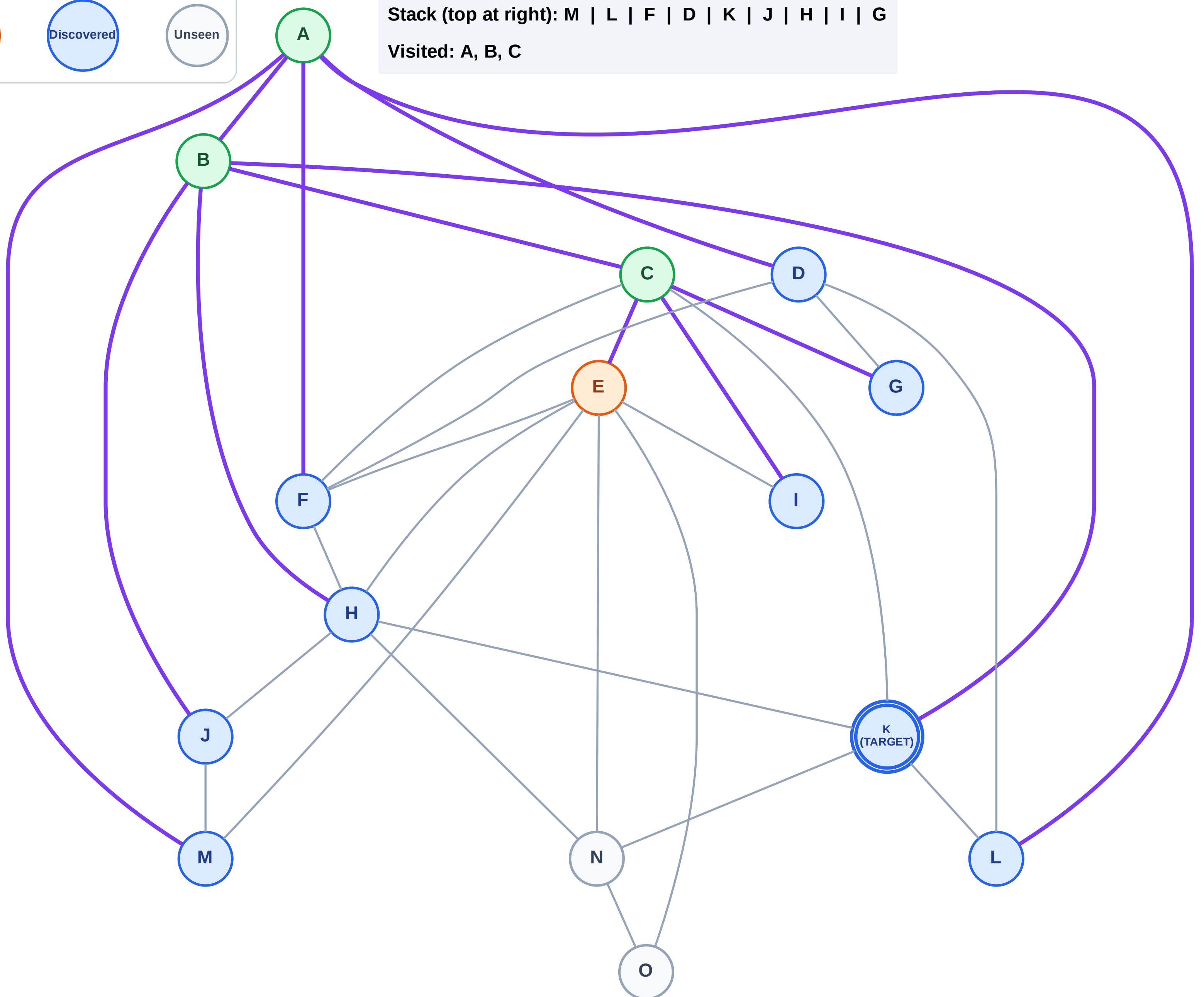


DFS Traversal

Step 8: Process node E

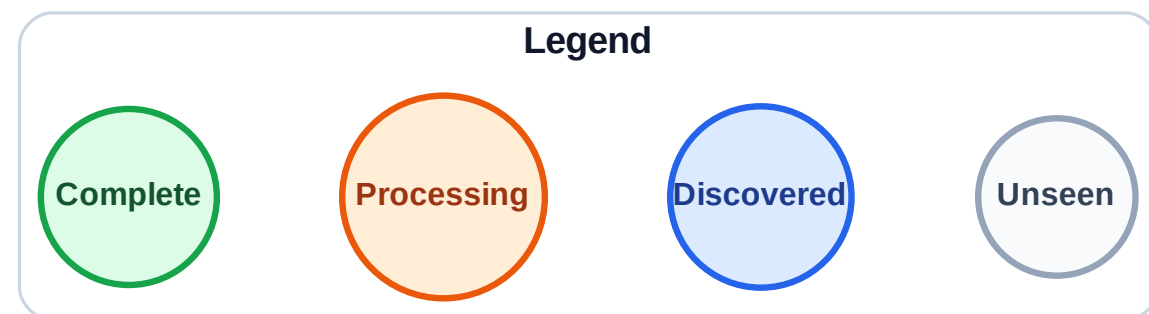


Stack (top at right): M | L | F | D | K | J | H | I | G
Visited: A, B, C



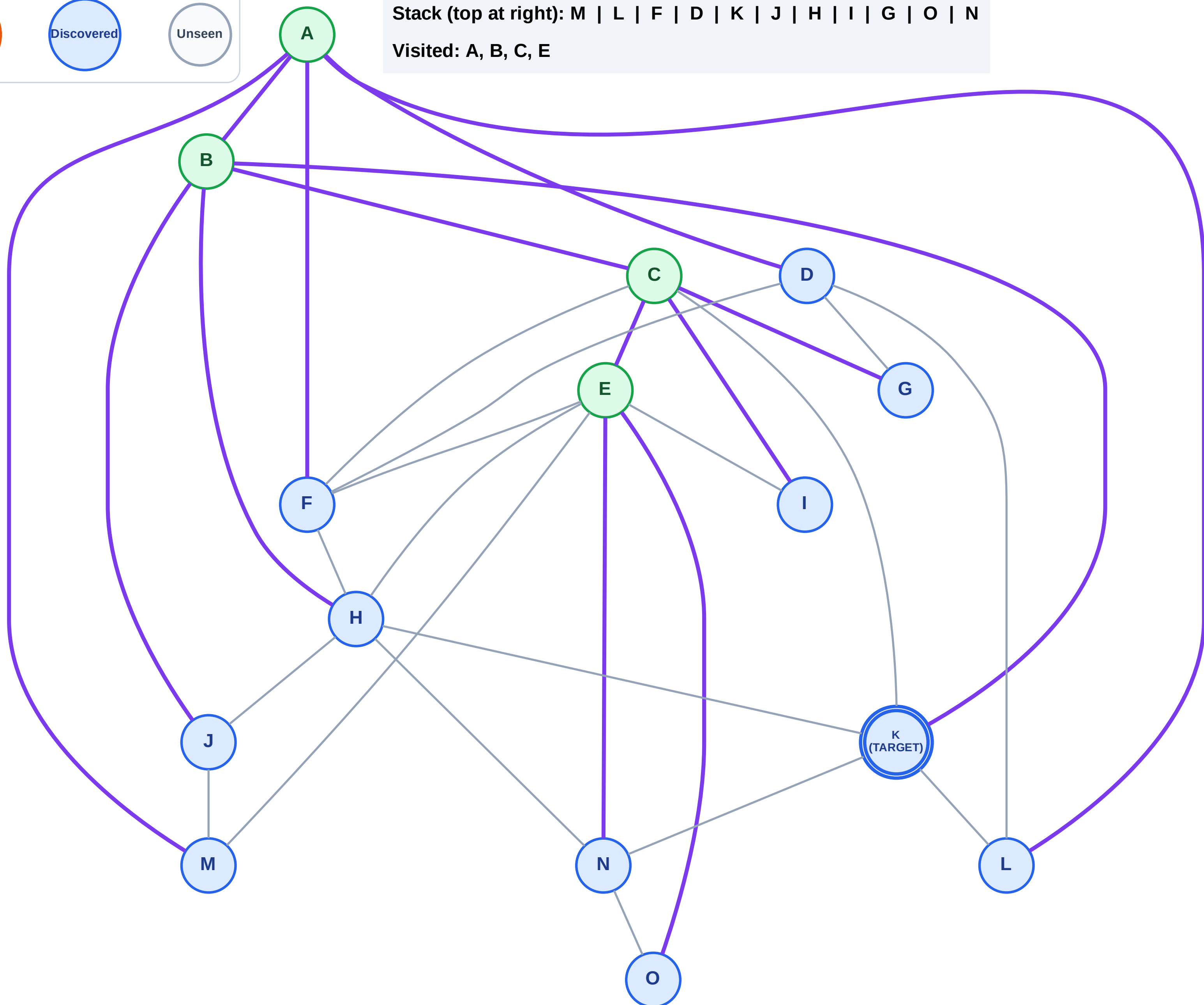
DFS Traversal

Step 9: Discover O, N from E



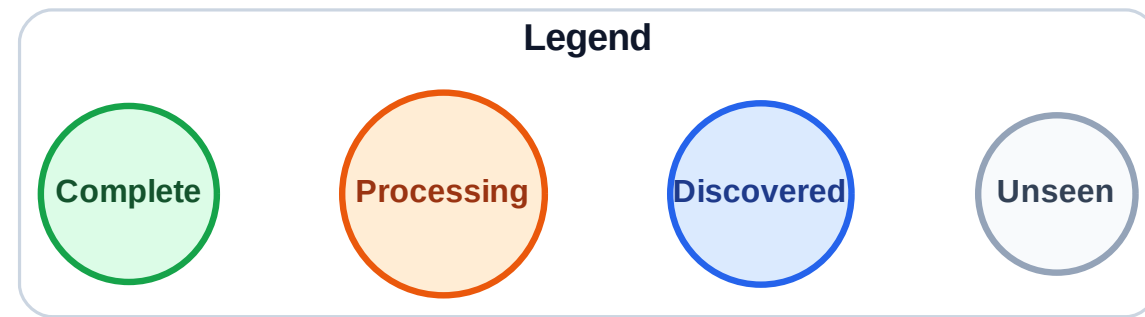
Stack (top at right): M | L | F | D | K | J | H | I | G | O | N

Visited: A, B, C, E



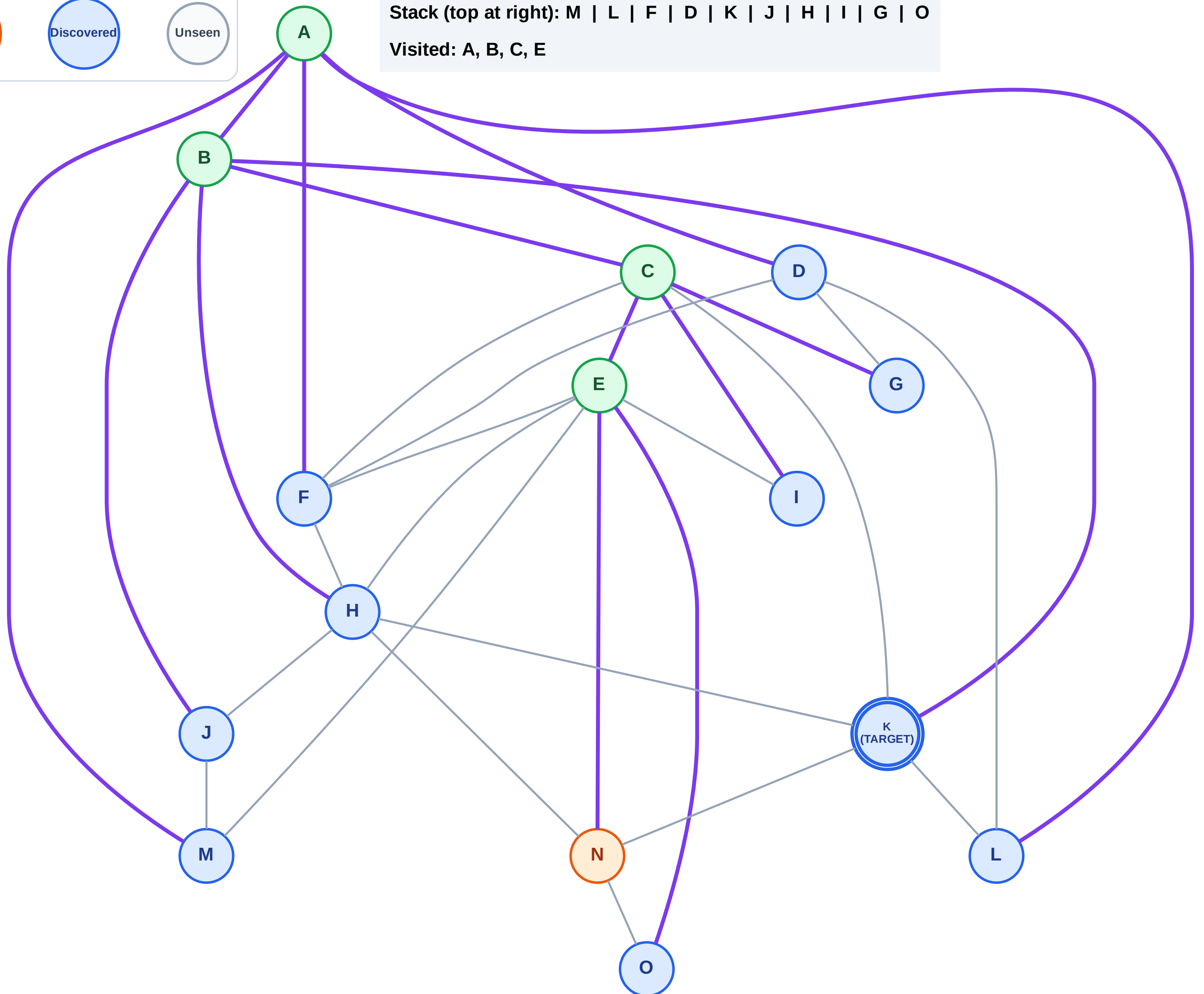
DFS Traversal

Step 10: Process node N



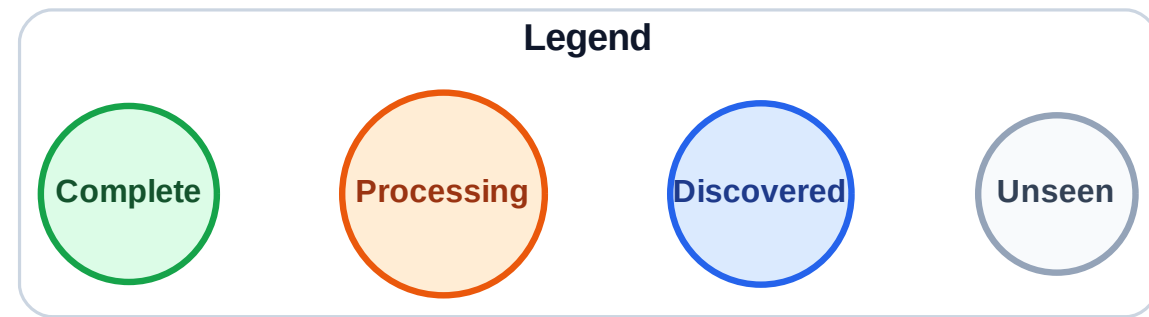
Stack (top at right): M | L | F | D | K | J | H | I | G | O

Visited: A, B, C, E



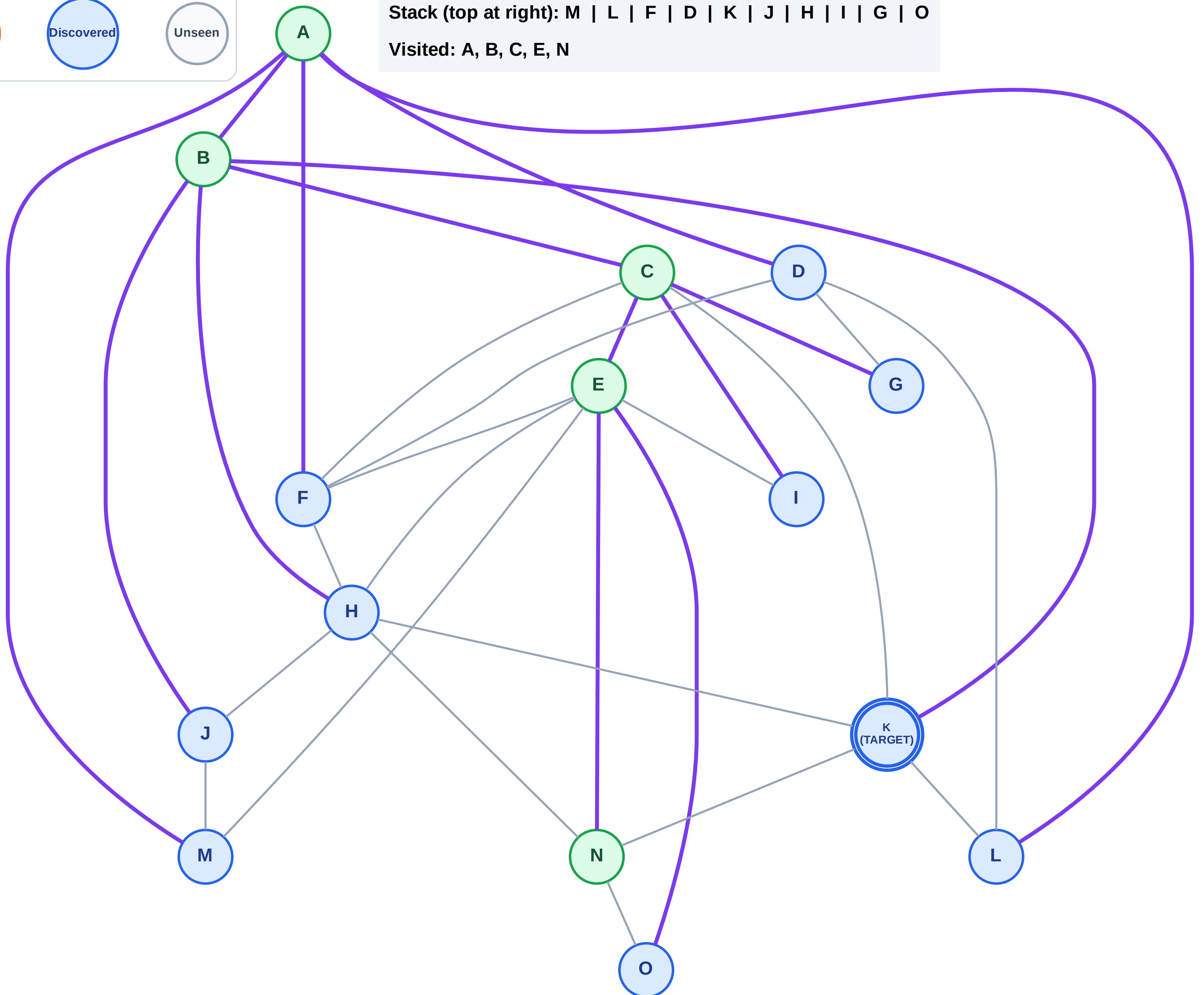
DFS Traversal

Step 11: No new neighbors from N



Stack (top at right): M | L | F | D | K | J | H | I | G | O

Visited: A, B, C, E, N



Step 12: Process node O



Complete

Complete



Processing

Processing



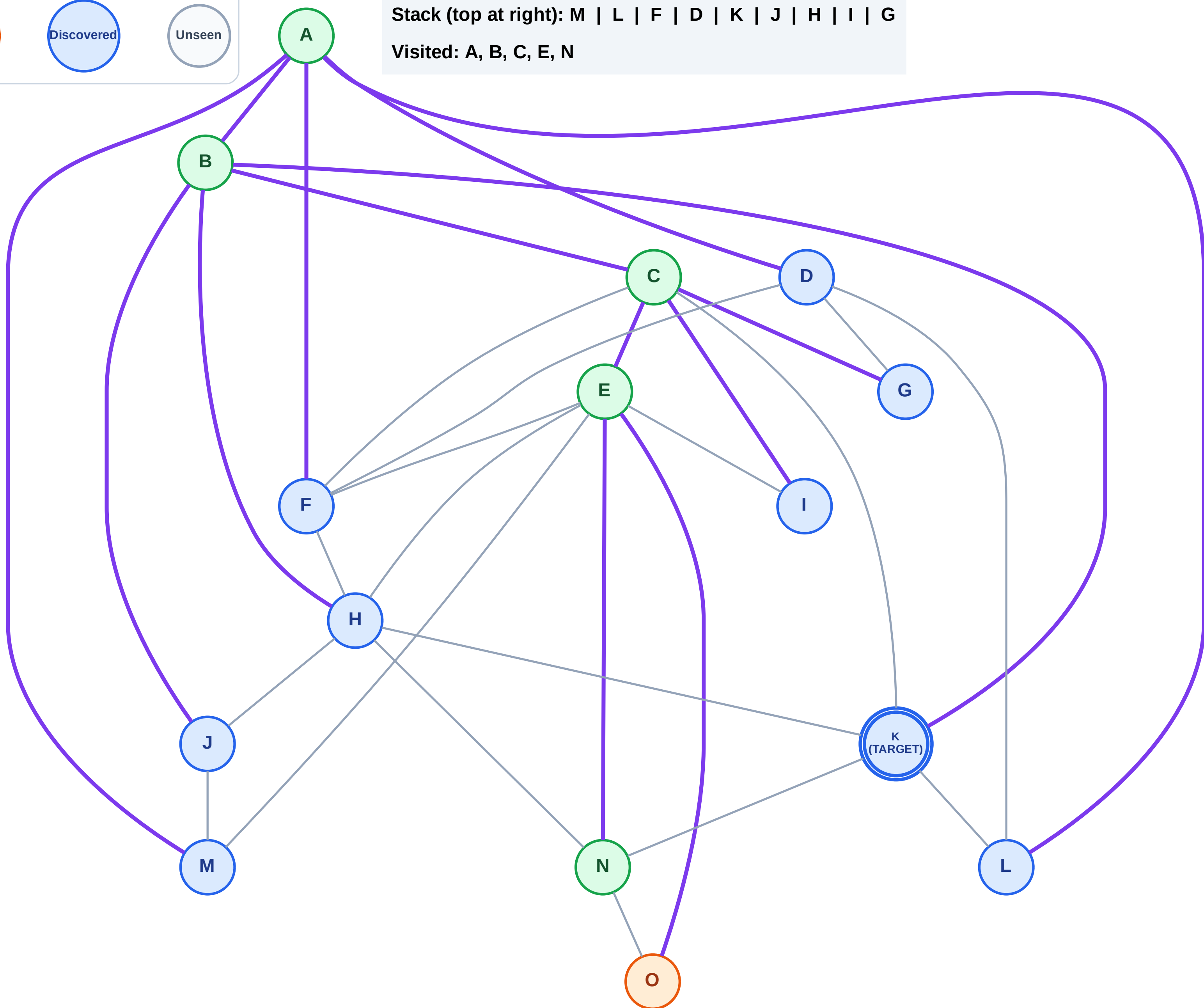
Discovered

Discovered

Unseen

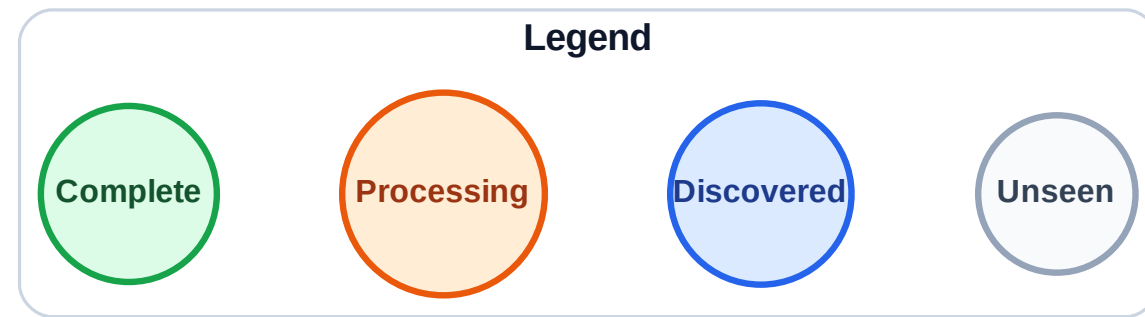
Unseen

Visited: A, B, C, E, N

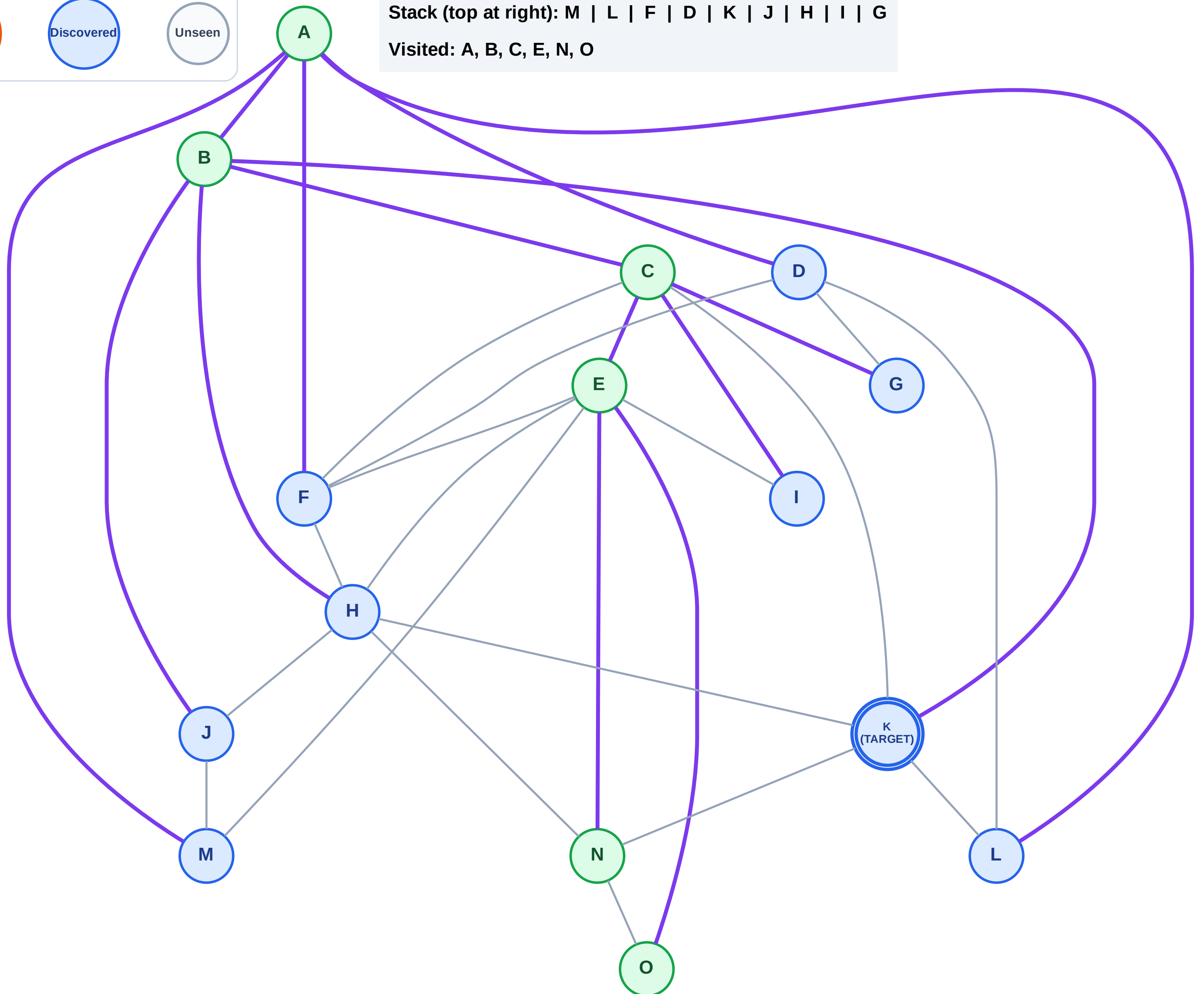


DFS Traversal

Step 13: No new neighbors from O

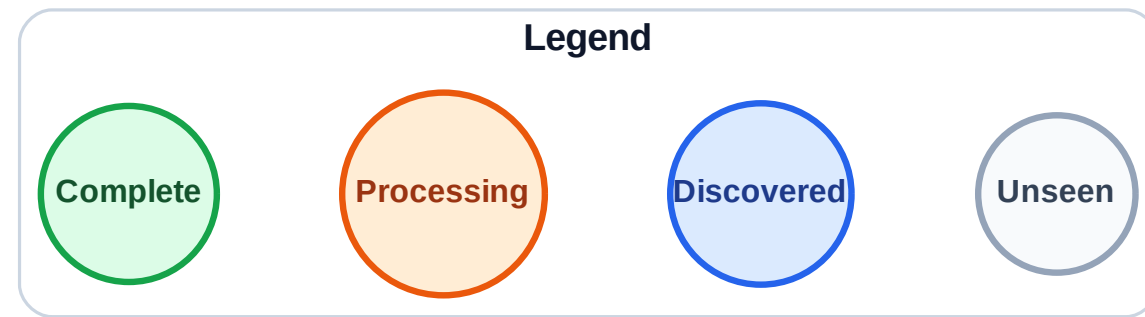


Stack (top at right): M | L | F | D | K | J | H | I | G
Visited: A, B, C, E, N, O



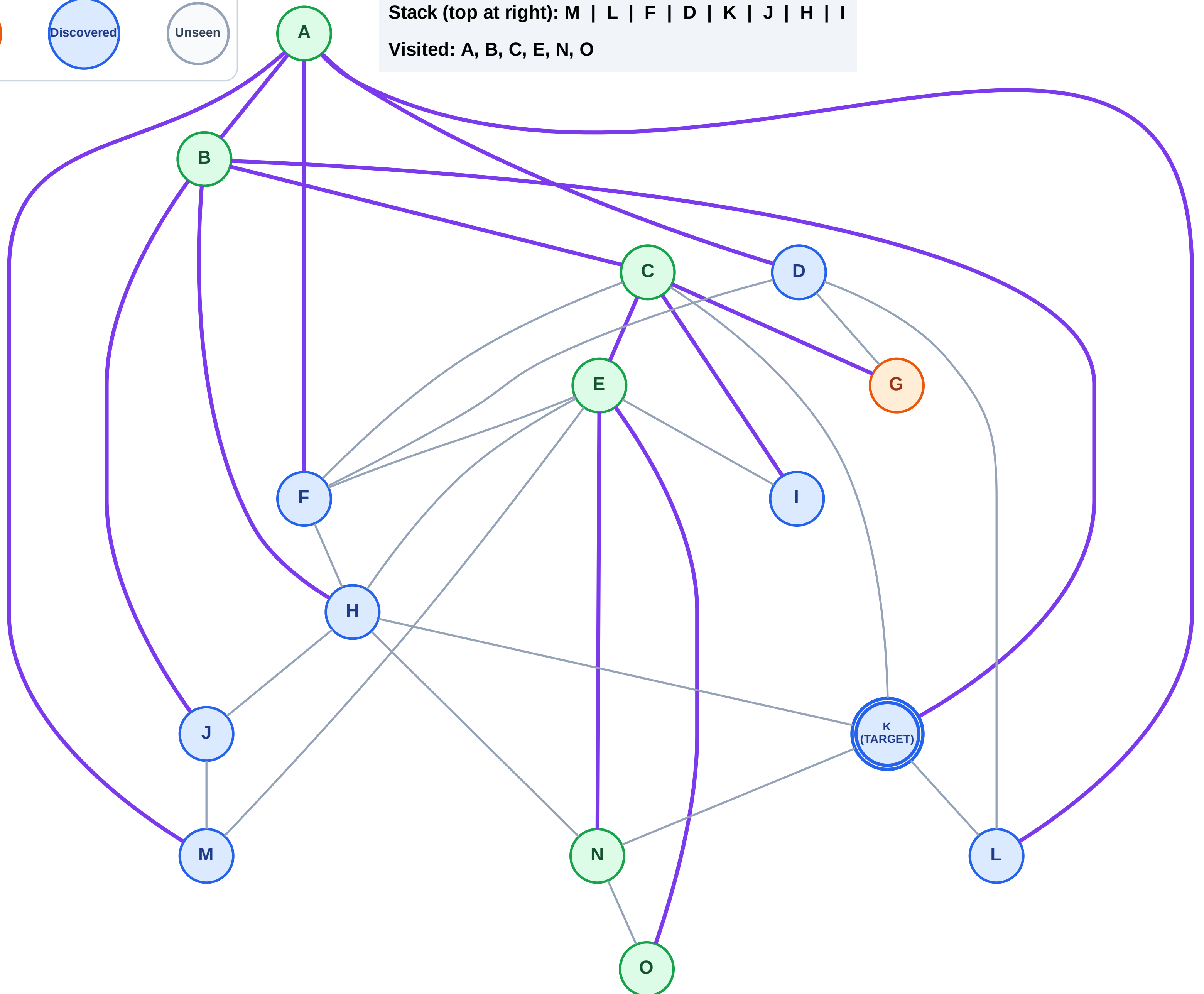
DFS Traversal

Step 14: Process node G



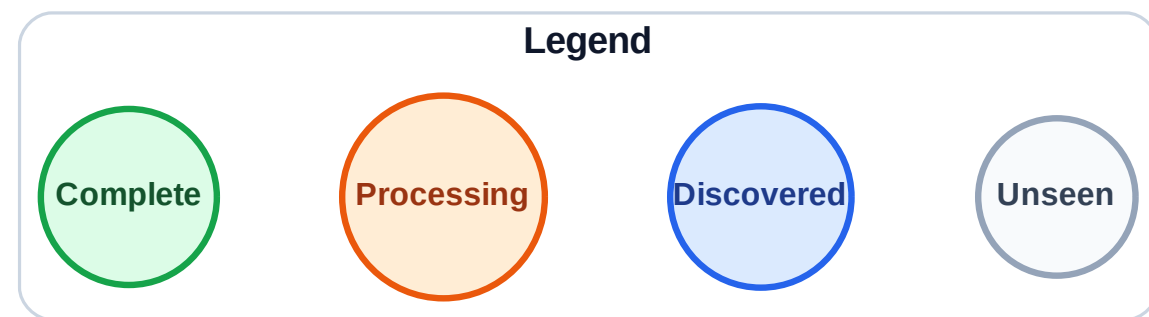
Stack (top at right): M | L | F | D | K | J | H | I

Visited: A, B, C, E, N, O



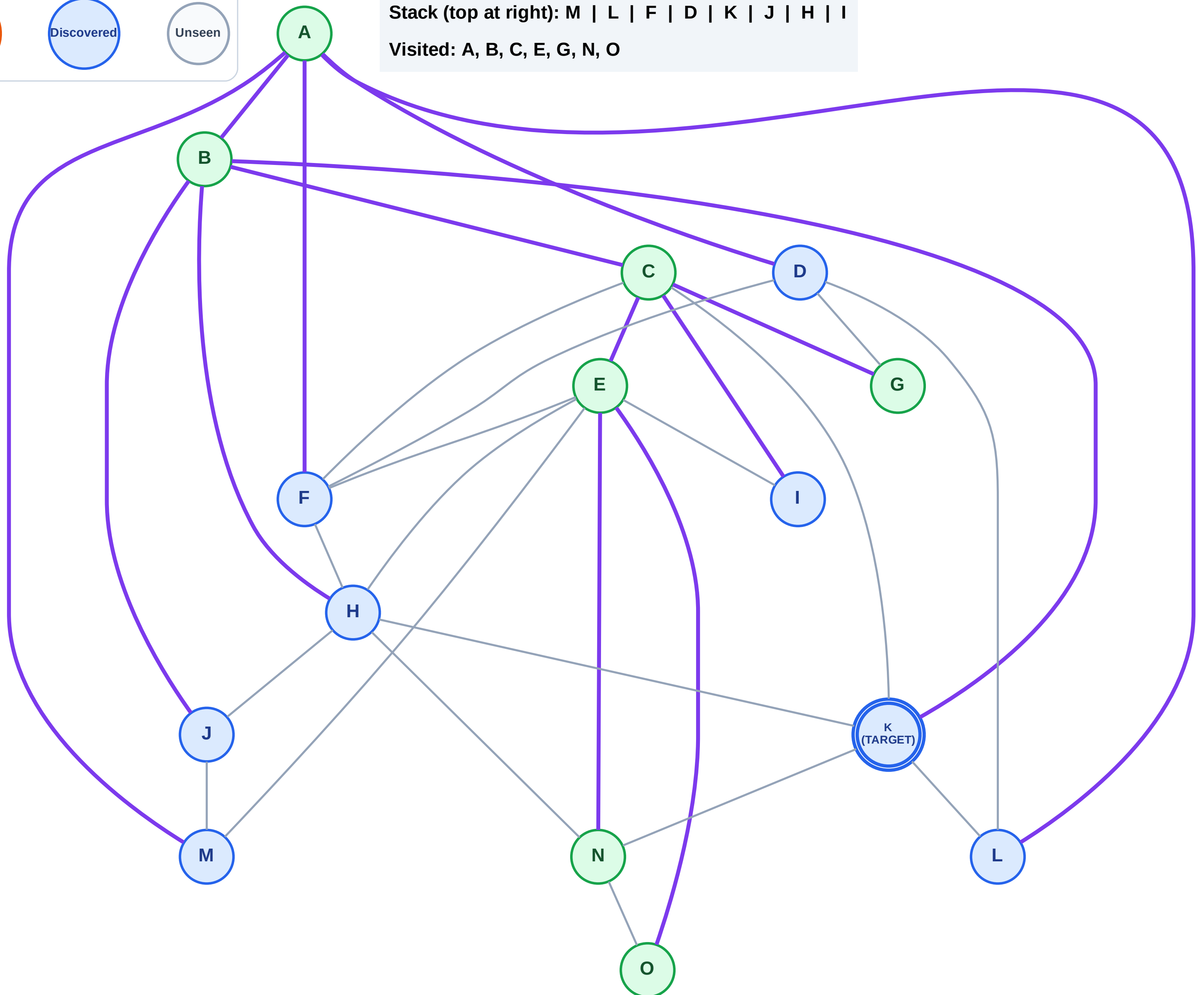
DFS Traversal

Step 15: No new neighbors from G



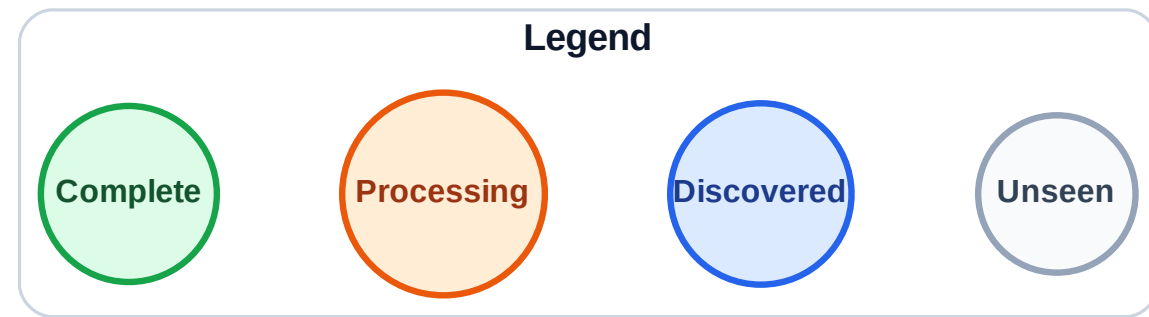
Stack (top at right): M | L | F | D | K | J | H | I

Visited: A, B, C, E, G, N, O



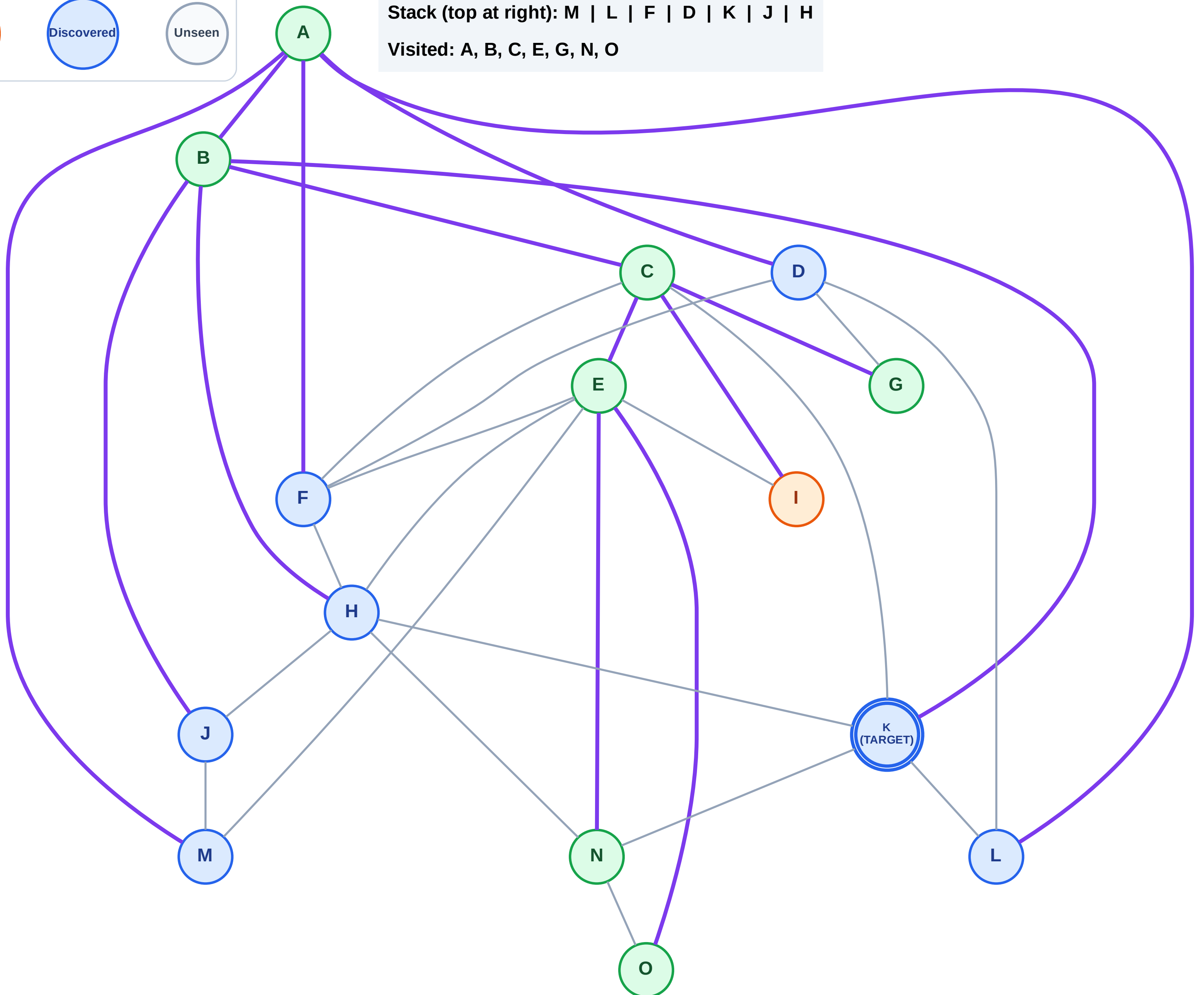
DFS Traversal

Step 16: Process node I



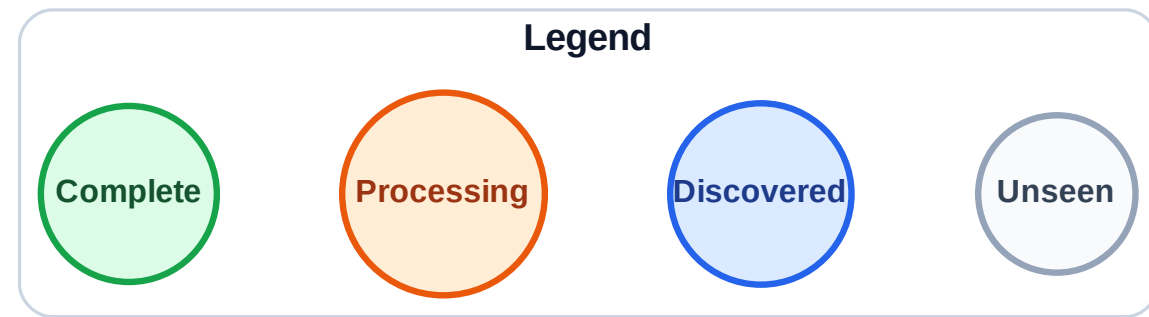
Stack (top at right): M | L | F | D | K | J | H

Visited: A, B, C, E, G, N, O



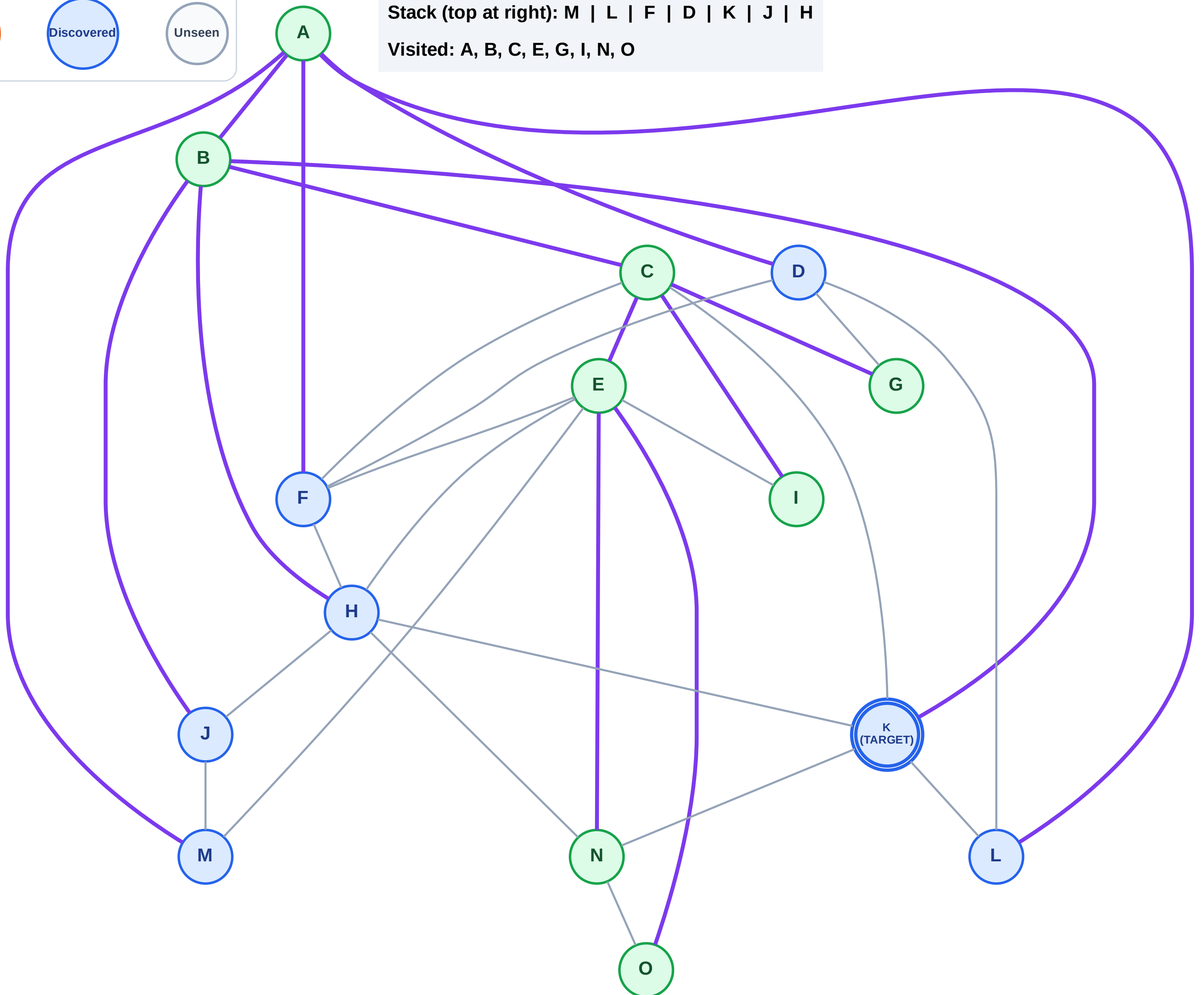
DFS Traversal

Step 17: No new neighbors from I



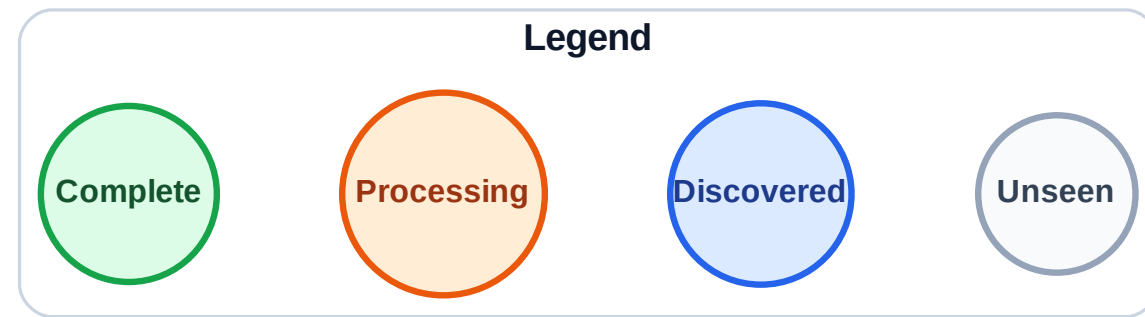
Stack (top at right): M | L | F | D | K | J | H

Visited: A, B, C, E, G, I, N, O



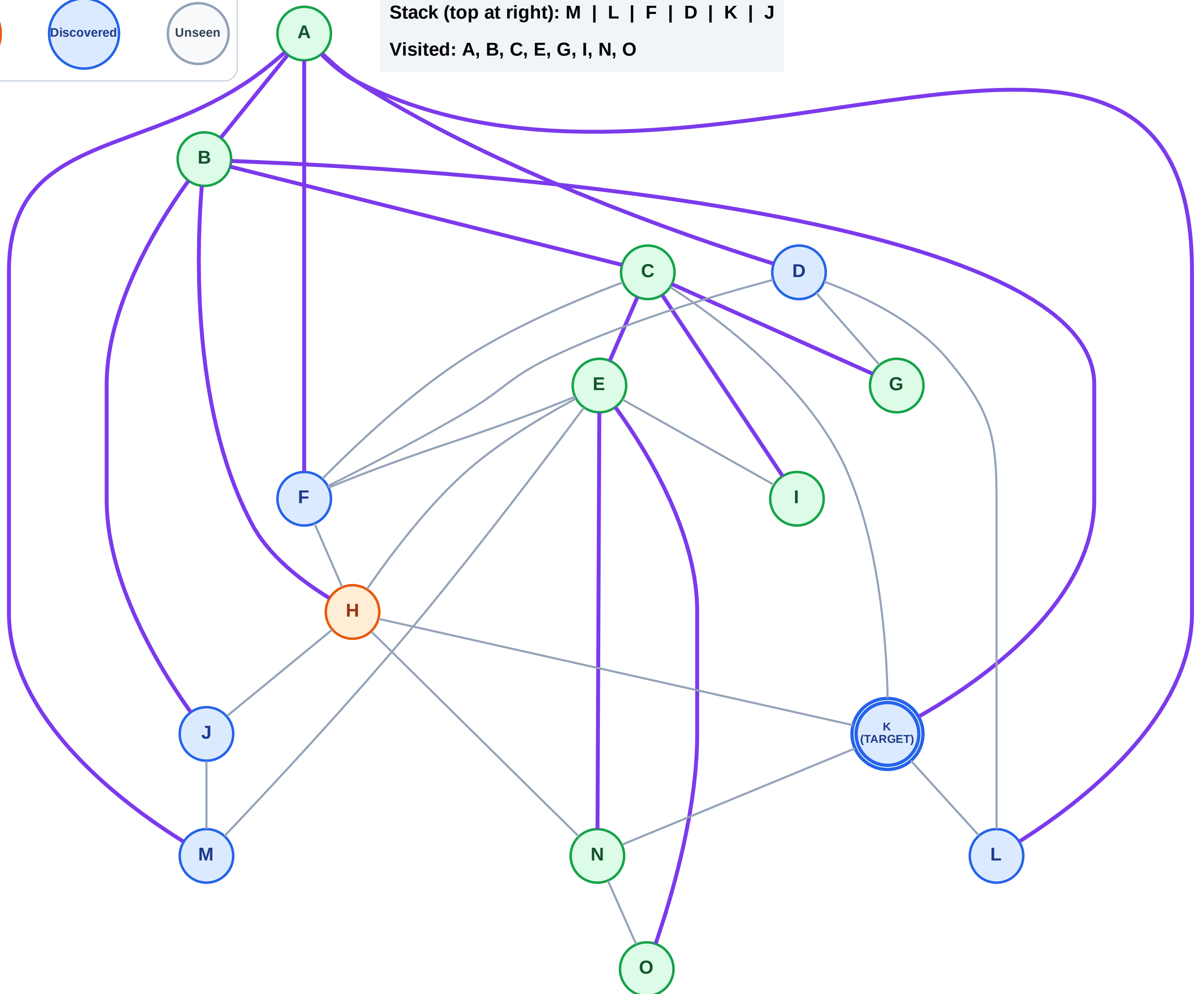
DFS Traversal

Step 18: Process node H



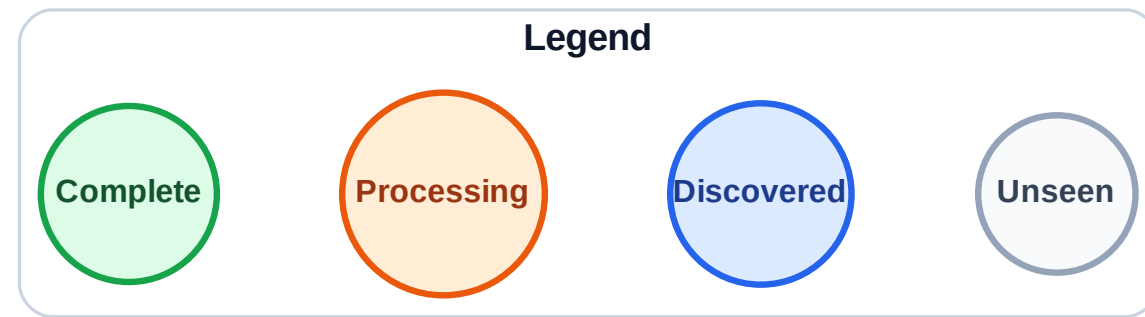
Stack (top at right): M | L | F | D | K | J

Visited: A, B, C, E, G, I, N, O



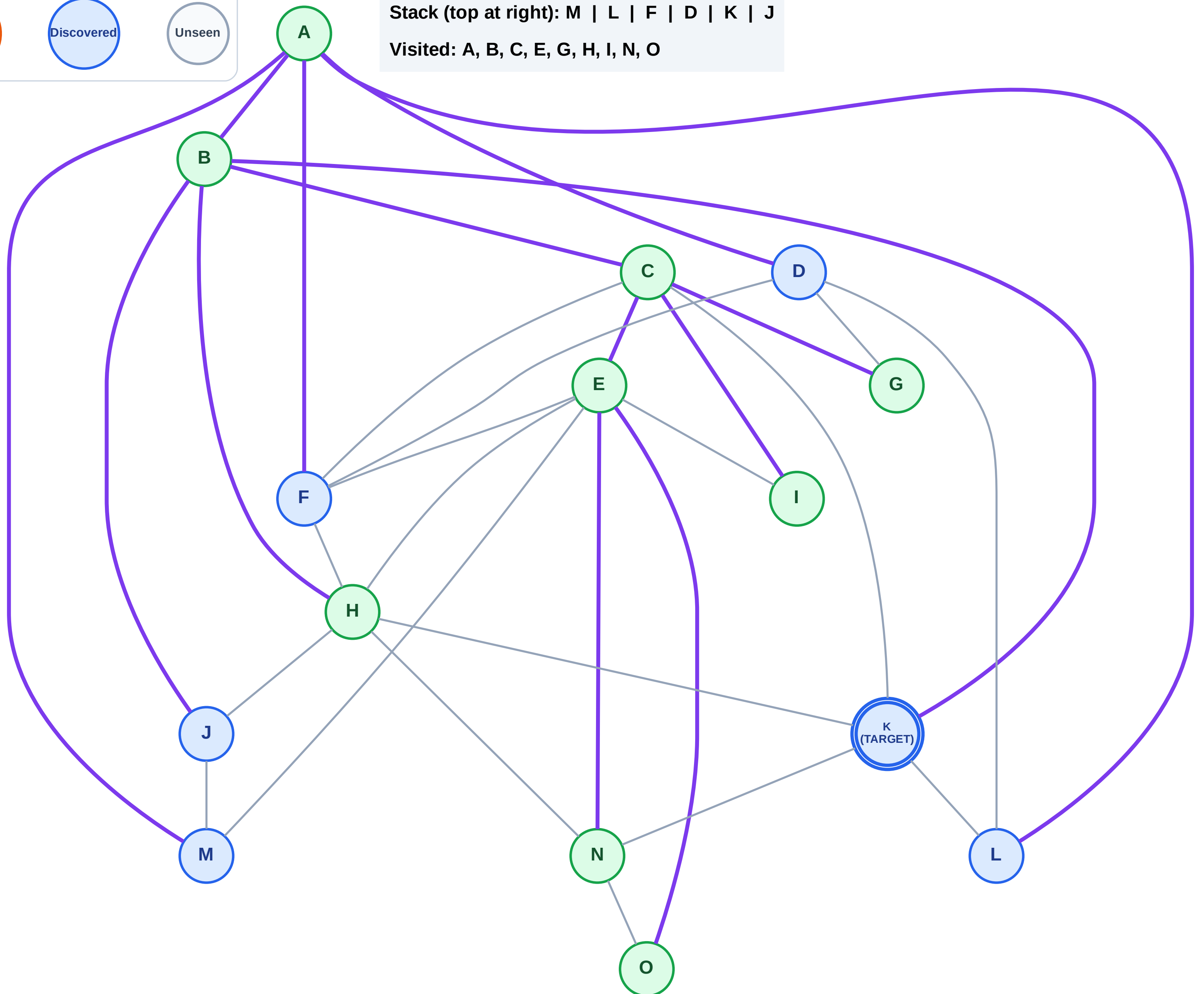
DFS Traversal

Step 19: No new neighbors from H



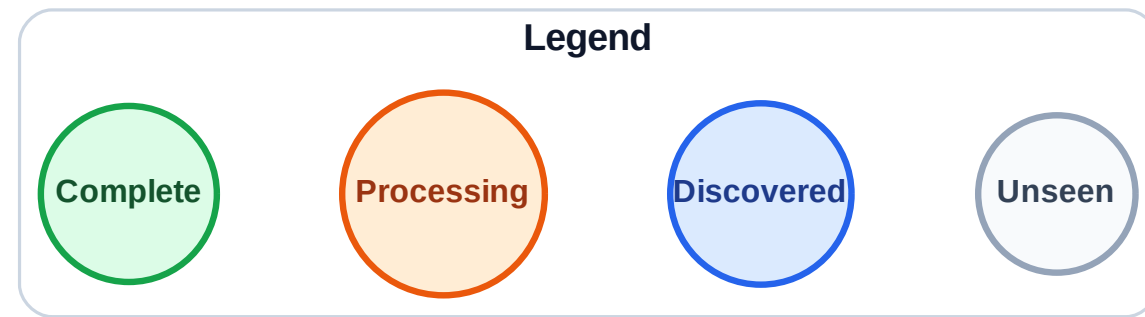
Stack (top at right): M | L | F | D | K | J

Visited: A, B, C, E, G, H, I, N, O



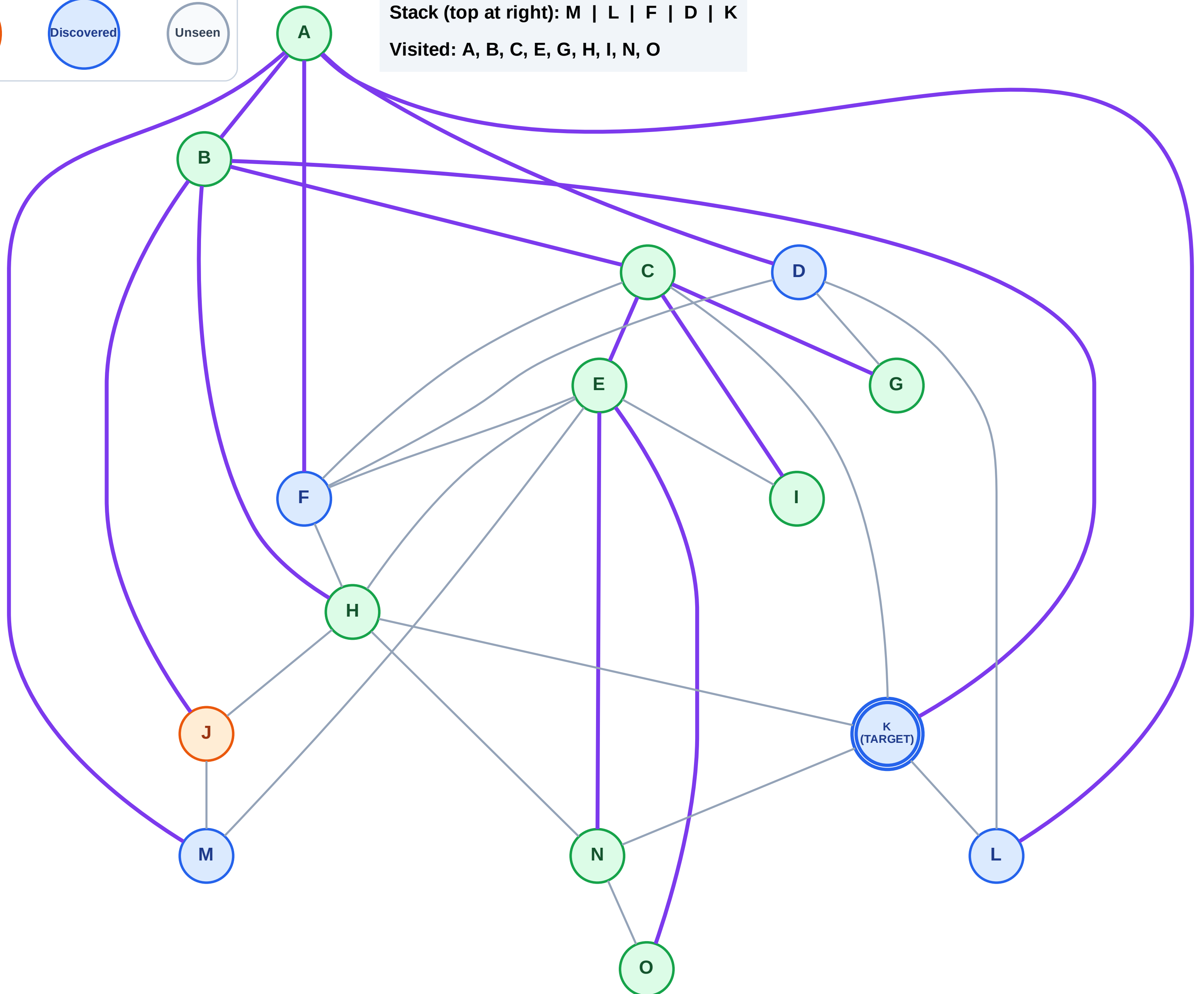
DFS Traversal

Step 20: Process node J



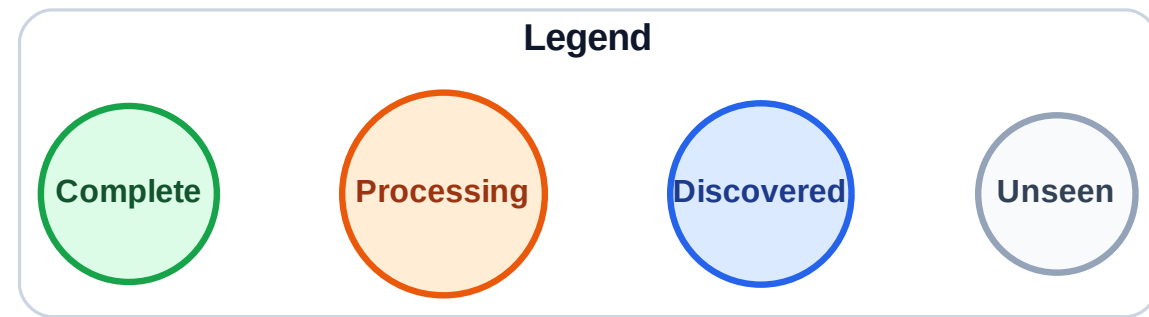
Stack (top at right): M | L | F | D | K

Visited: A, B, C, E, G, H, I, N, O



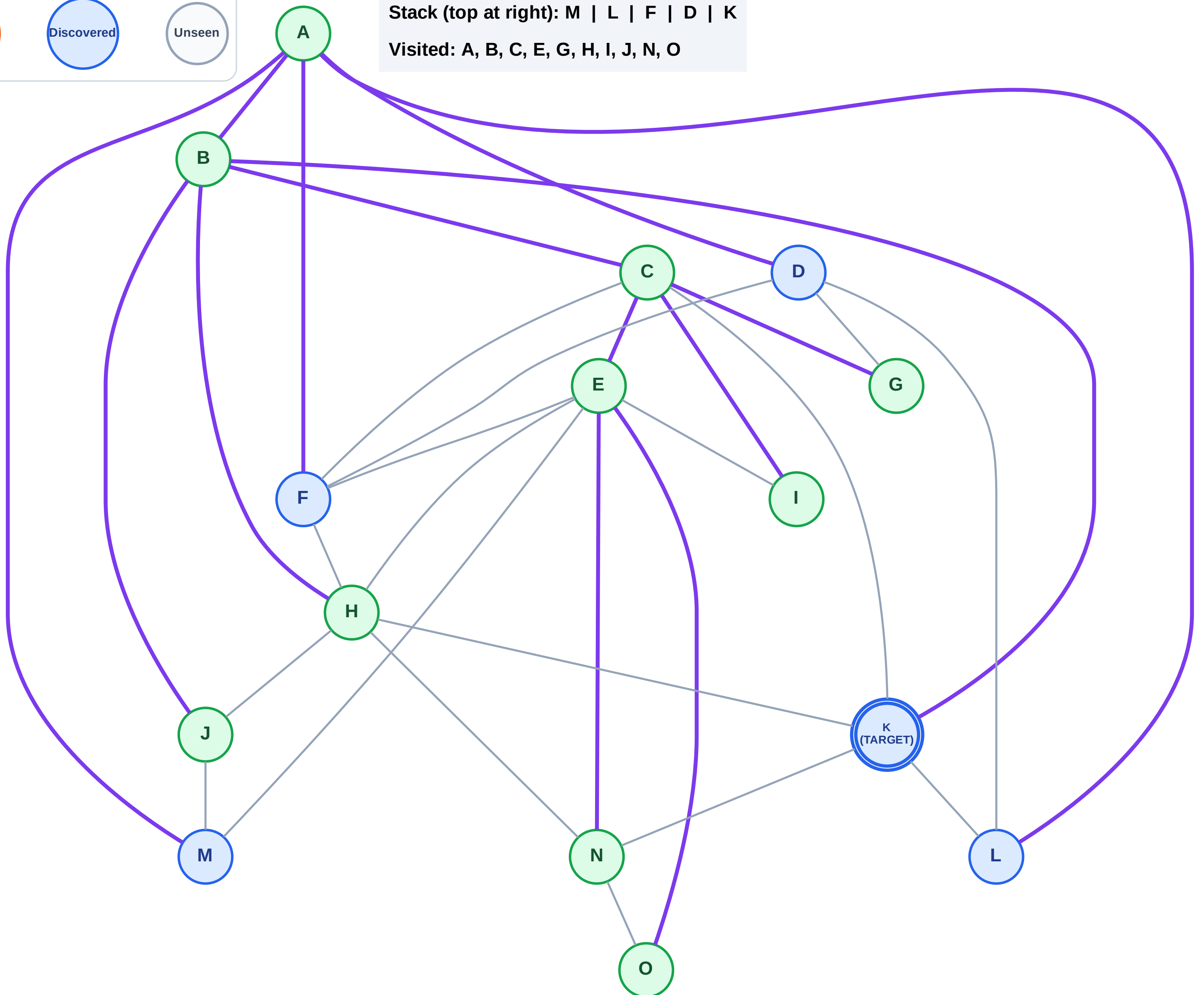
DFS Traversal

Step 21: No new neighbors from J



Stack (top at right): M | L | F | D | K

Visited: A, B, C, E, G, H, I, J, N, O



DFS Traversal

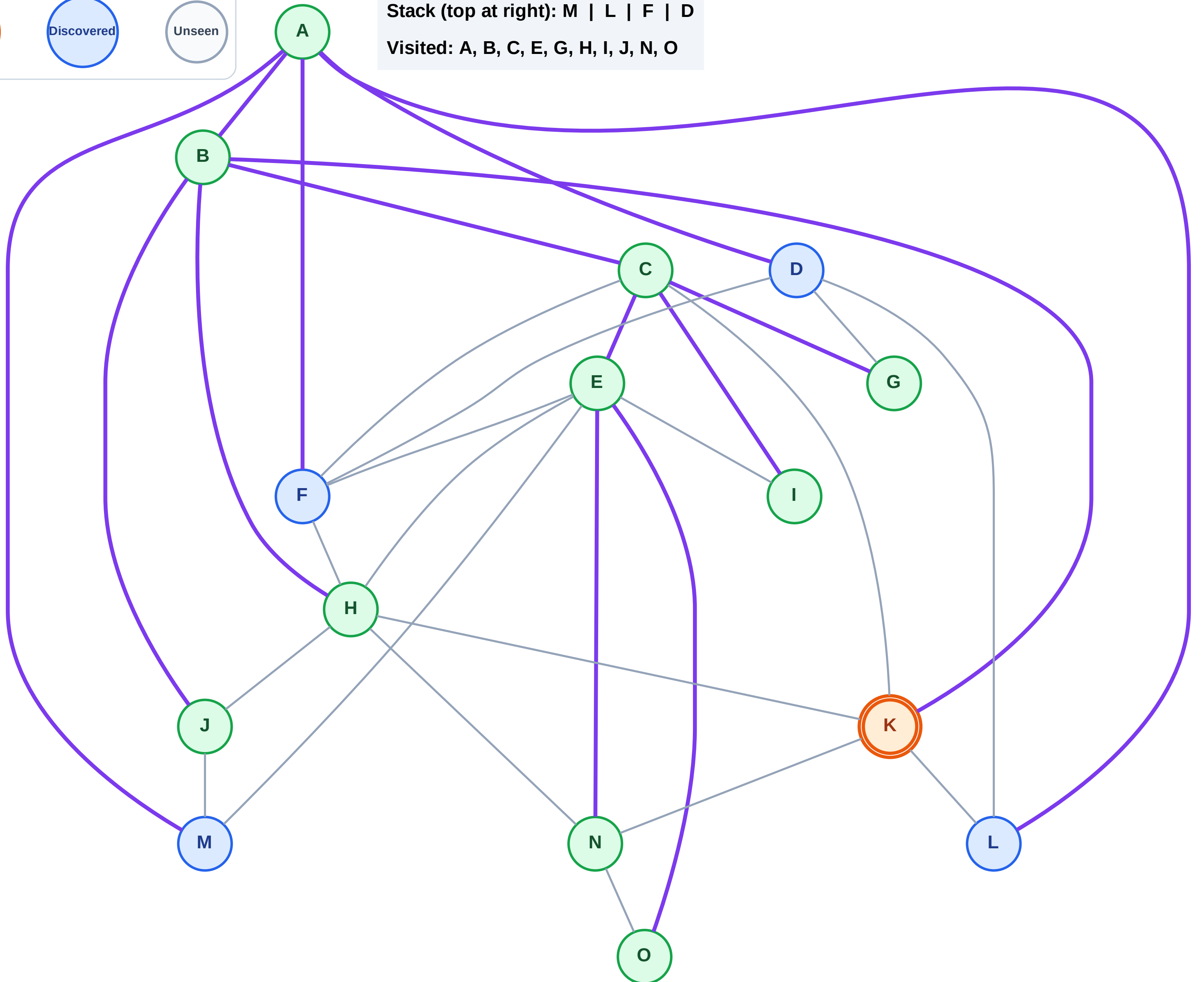
Step 22: Process node K

Legend



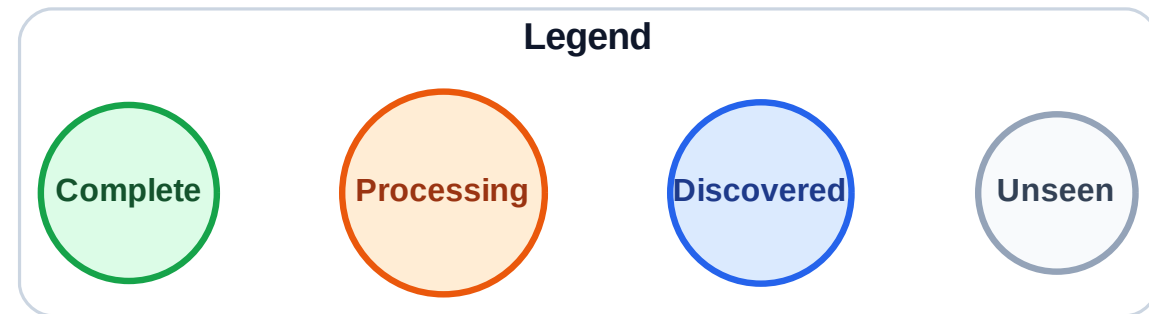
Stack (top at right): M | L | F | D

Visited: A, B, C, E, G, H, I, J, N, O



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K



Stack (top at right): M | L | F | D
Visited: A, B, C, E, G, H, I, J, K, N, O

